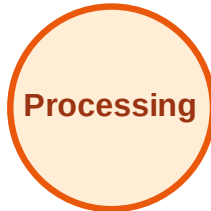


BFS Traversal

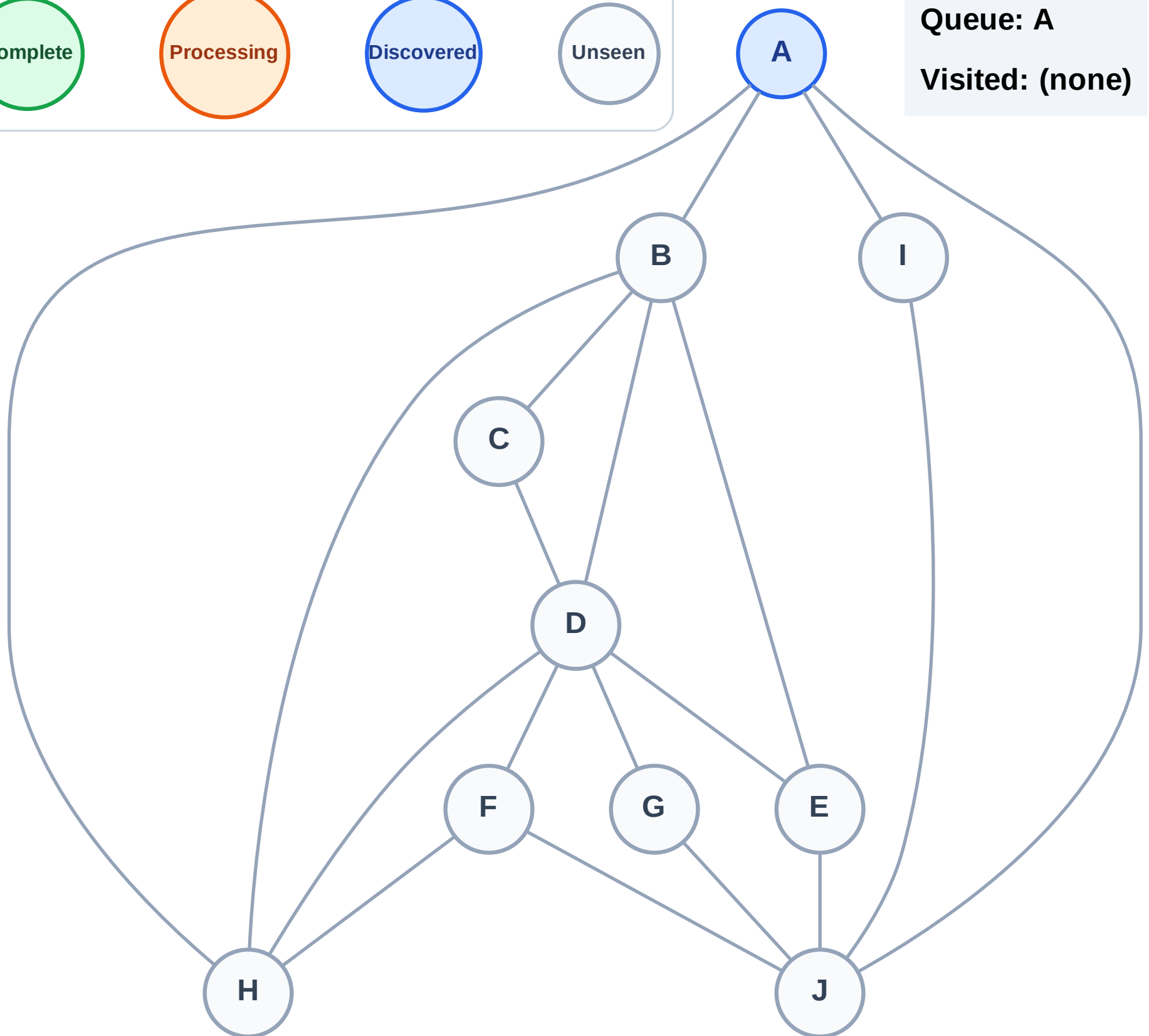
Step 1: Start at A

Legend



Queue: A

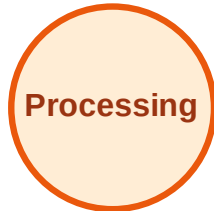
Visited: (none)



BFS Traversal

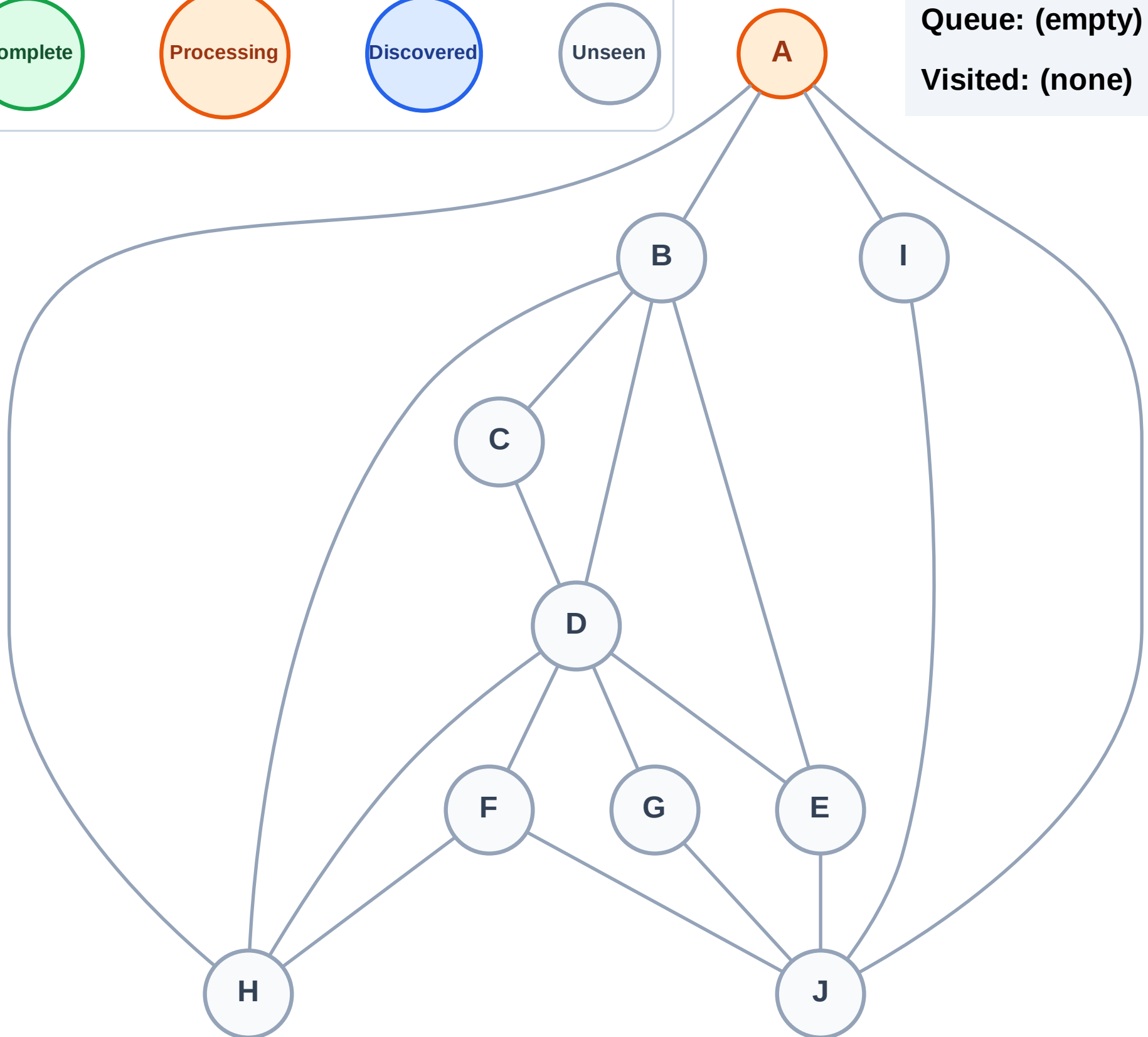
Step 2: Process node A

Legend



Queue: (empty)

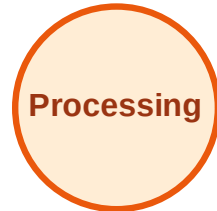
Visited: (none)



BFS Traversal

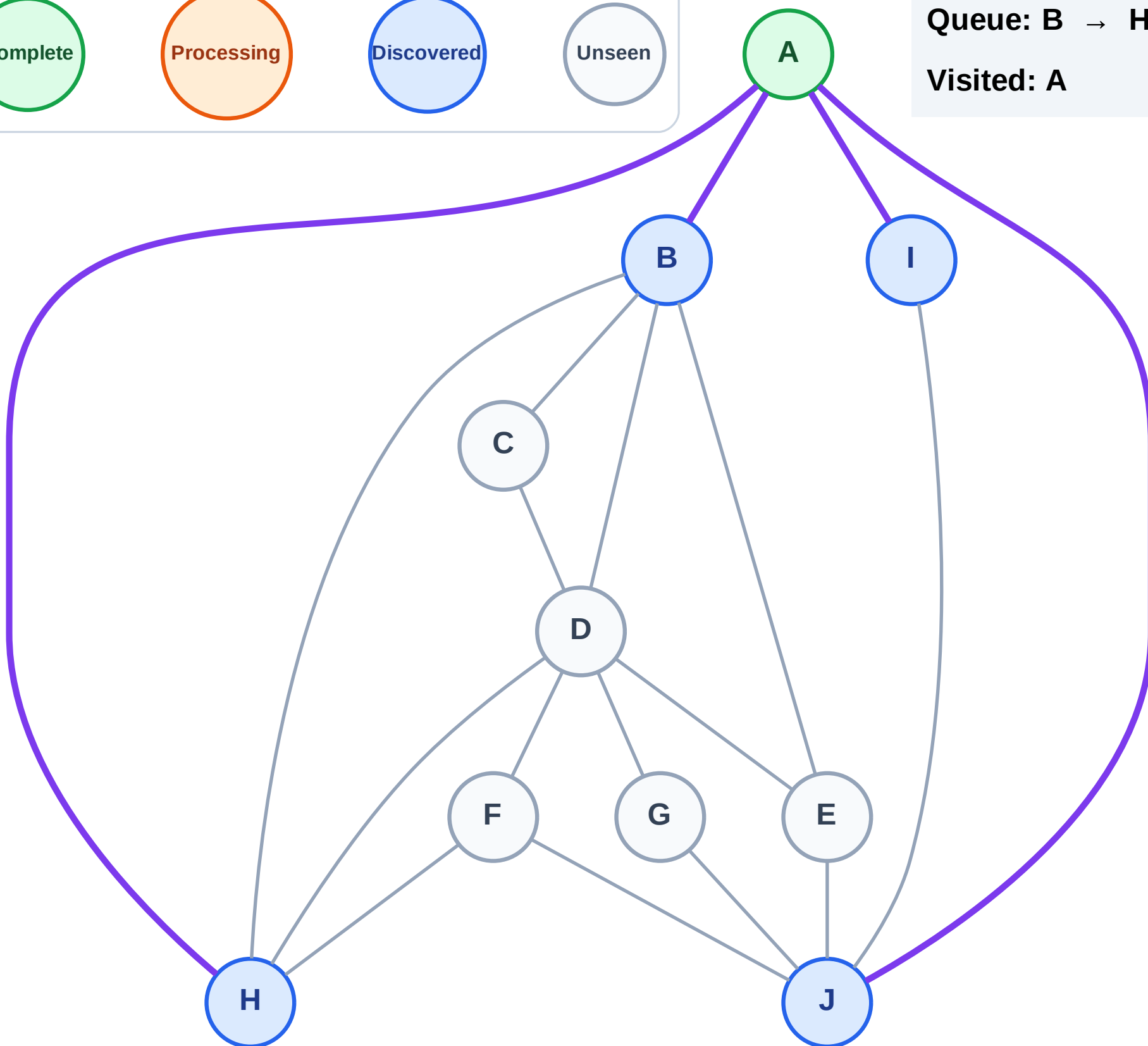
Step 3: Discover B, H, I, J from A

Legend



Queue: B → H → I → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

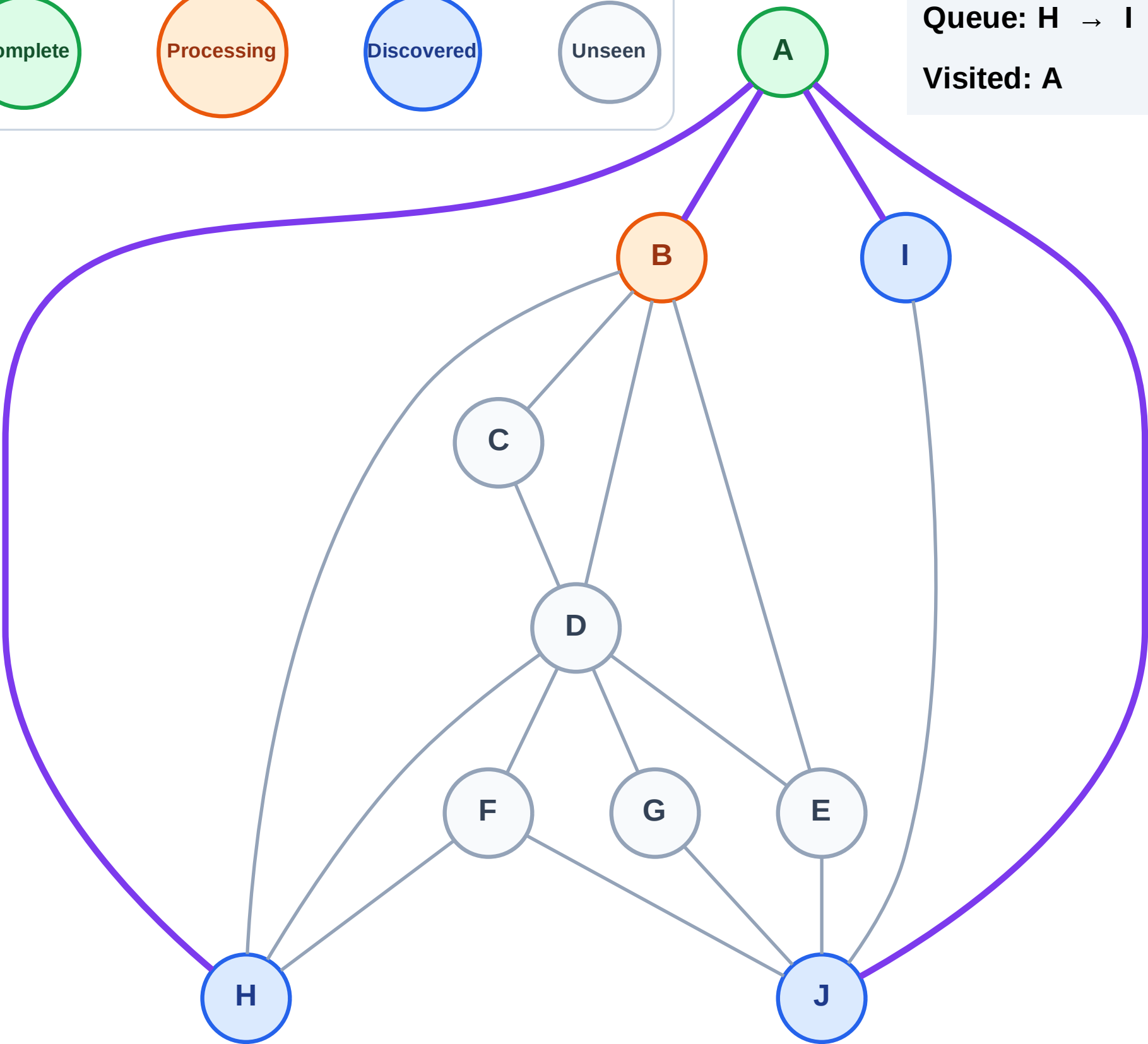
Processing

Discovered

Unseen

Queue: H → I → J

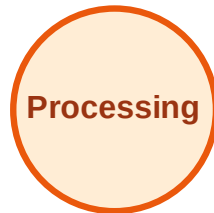
Visited: A



BFS Traversal

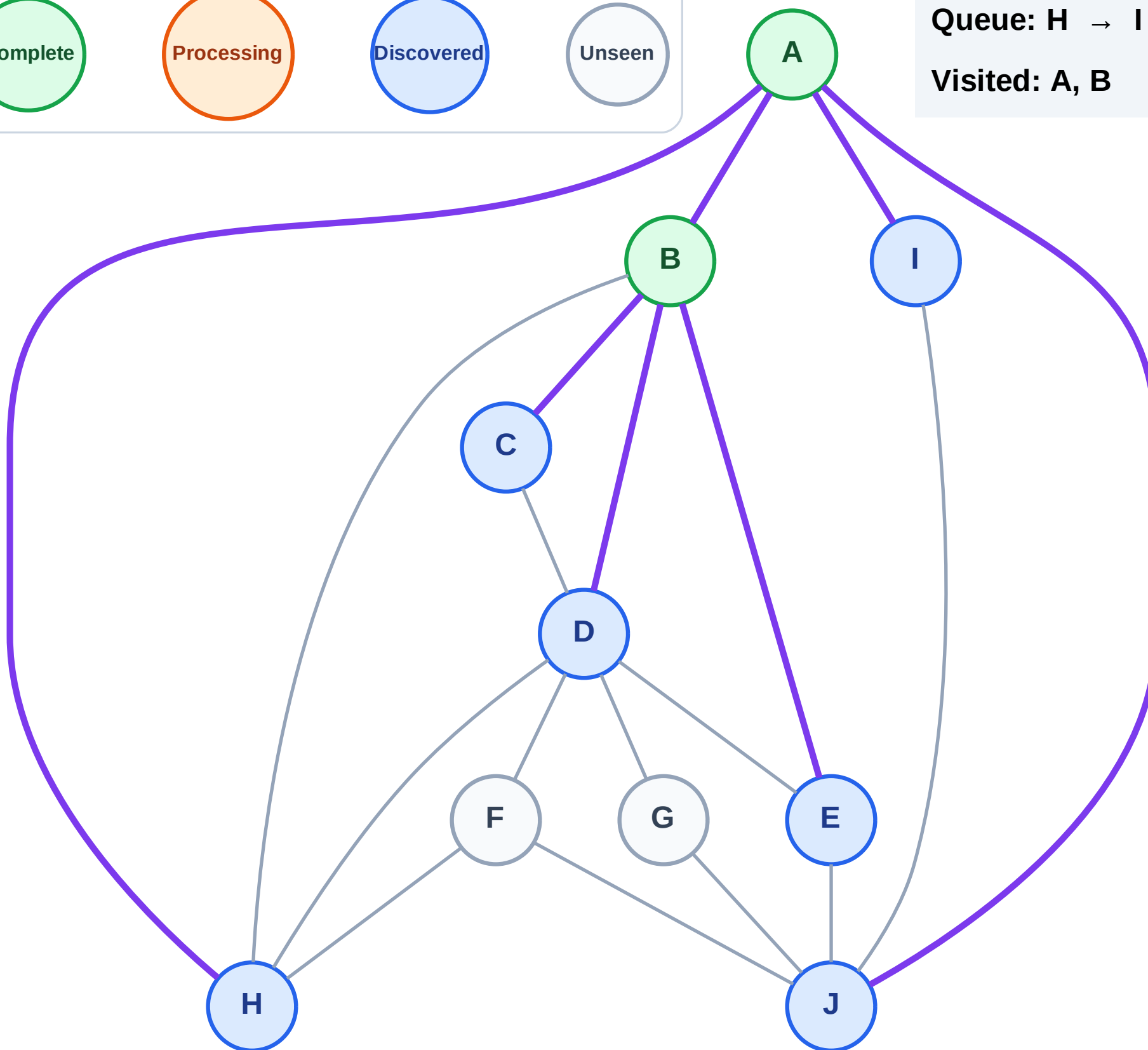
Step 5: Discover C, D, E from B

Legend



Queue: H → I → J → C → D → E

Visited: A, B



BFS Traversal

Step 6: Process node H

Legend

Complete

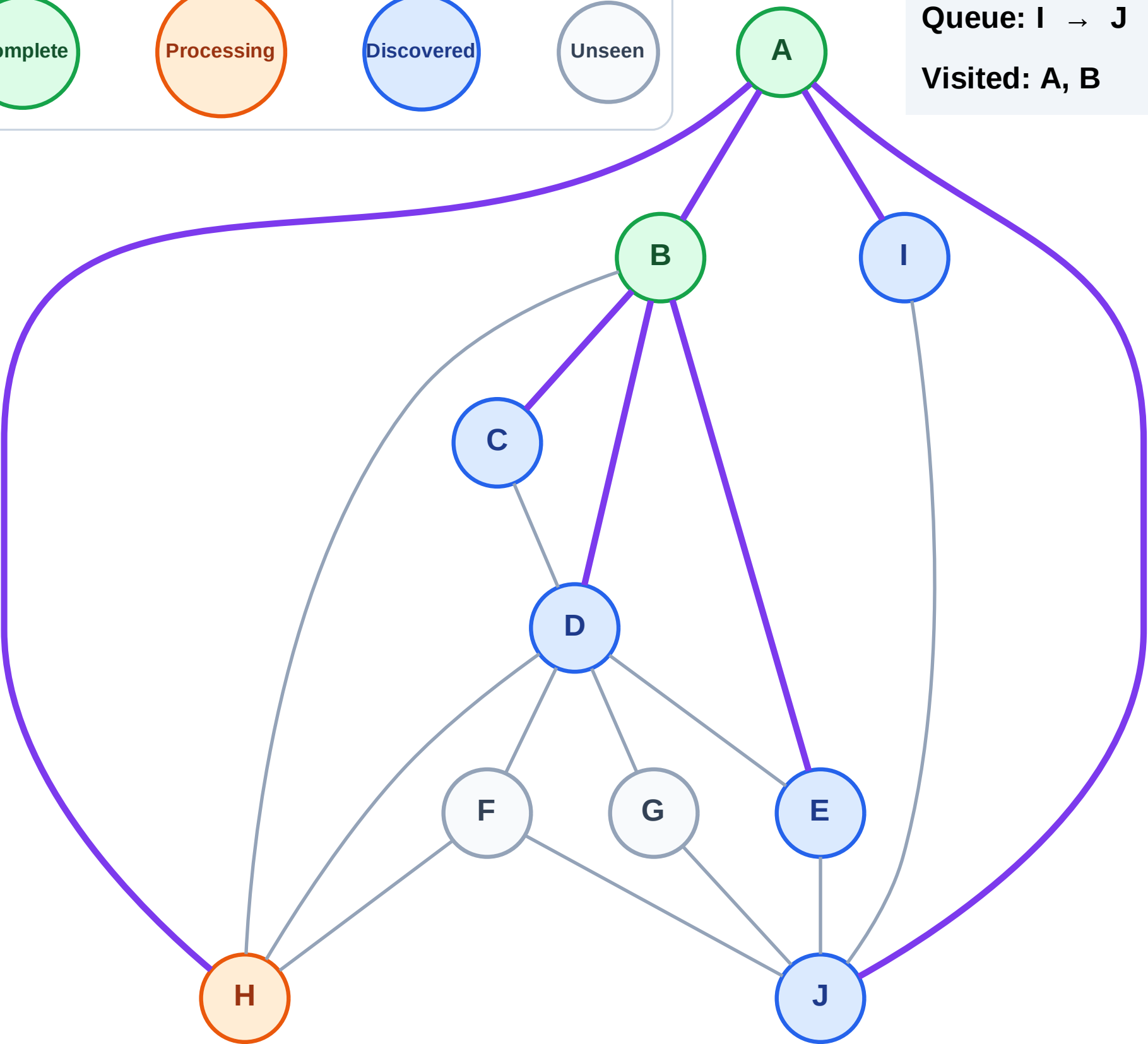
Processing

Discovered

Unseen

Queue: I → J → C → D → E

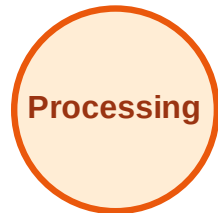
Visited: A, B



BFS Traversal

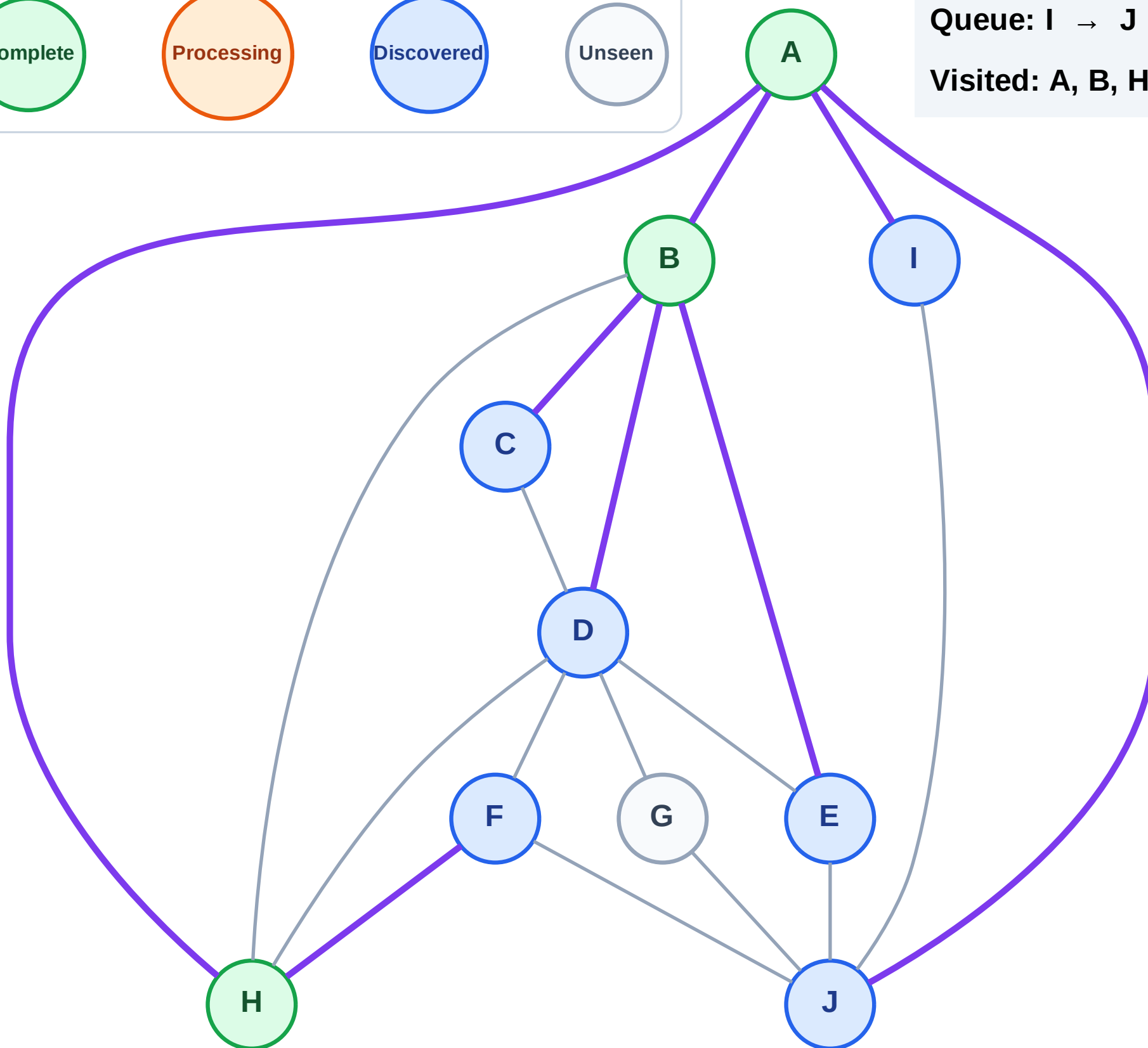
Step 7: Discover F from H

Legend



Queue: I → J → C → D → E → F

Visited: A, B, H



BFS Traversal

Step 8: Process node I

Legend

Complete

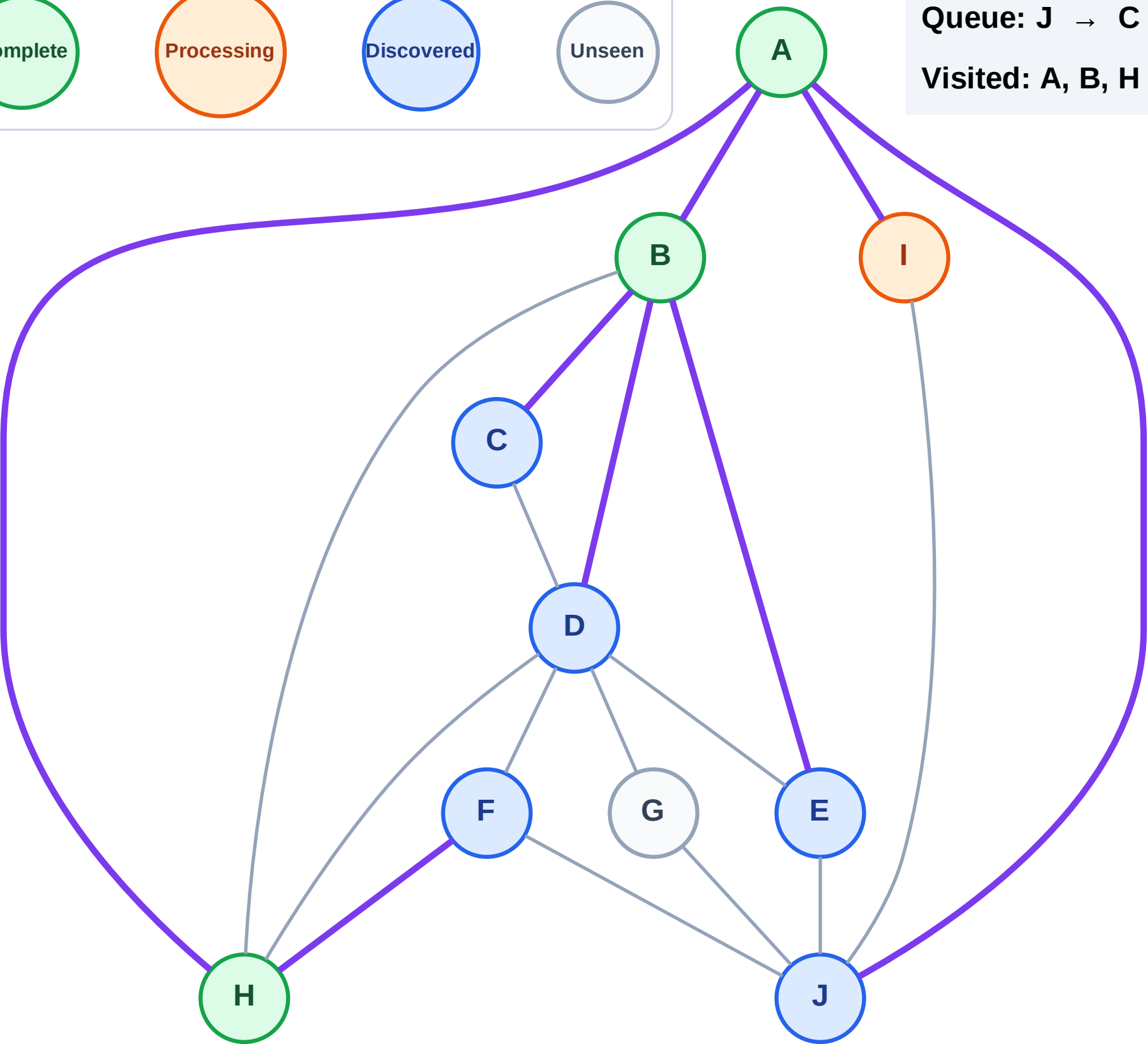
Processing

Discovered

Unseen

Queue: J → C → D → E → F

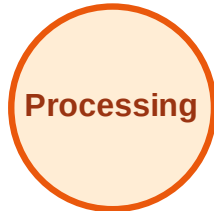
Visited: A, B, H



BFS Traversal

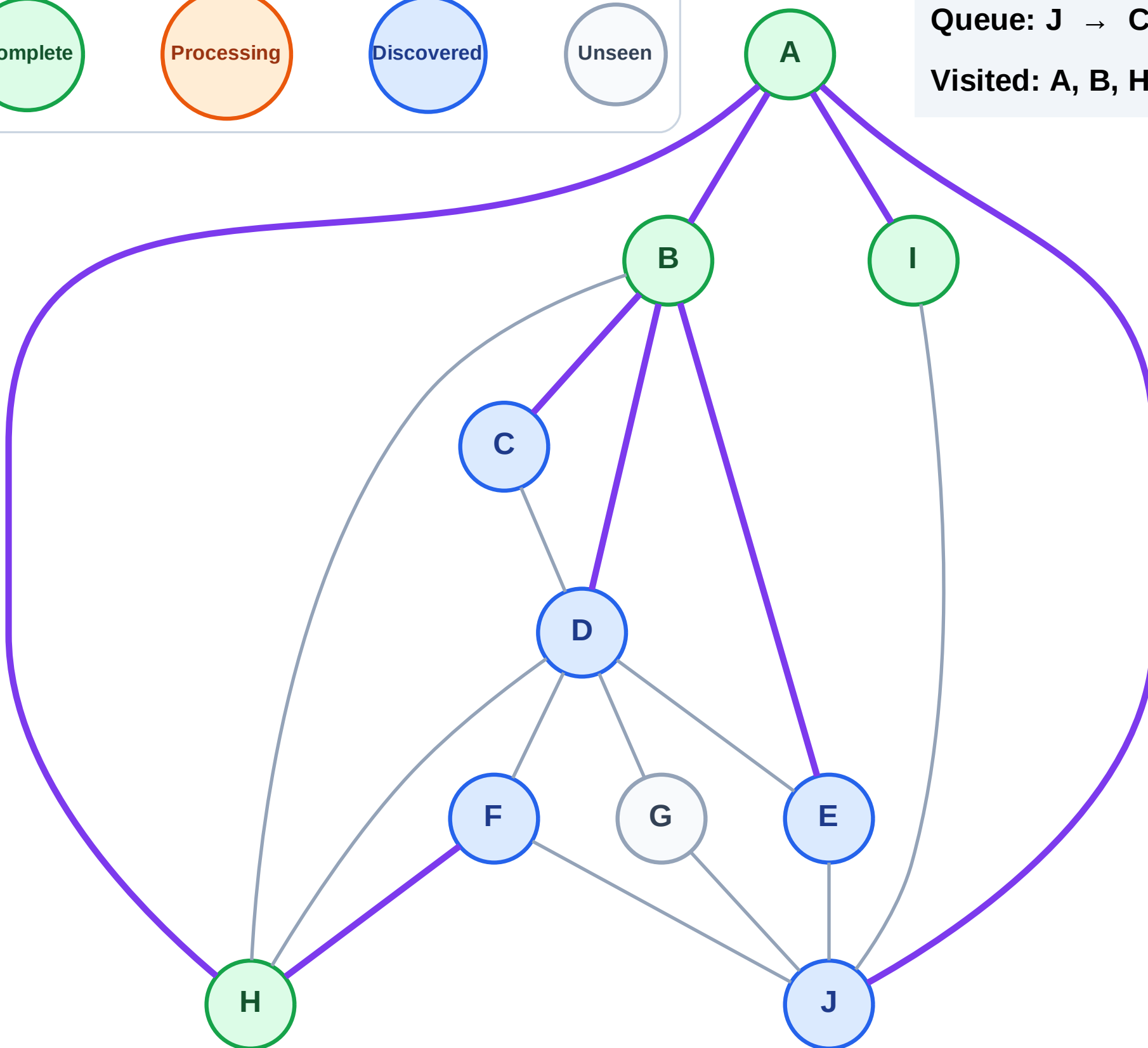
Step 9: No new neighbors from I

Legend



Queue: J → C → D → E → F

Visited: A, B, H, I



BFS Traversal

Step 10: Process node J

Legend

Complete

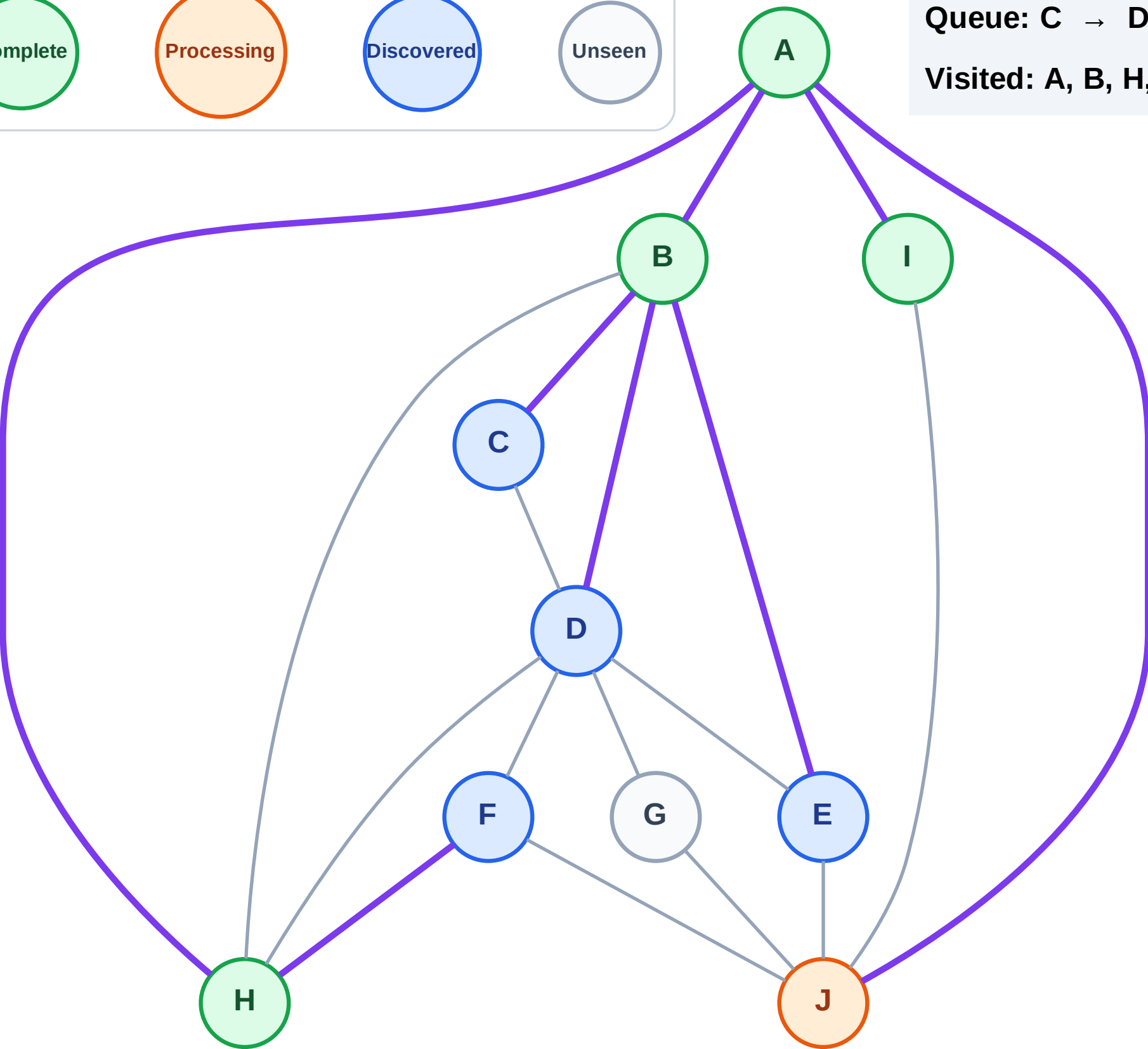
Processing

Discovered

Unseen

Queue: C → D → E → F

Visited: A, B, H, I



BFS Traversal

Step 11: Discover G from J

Legend

Complete

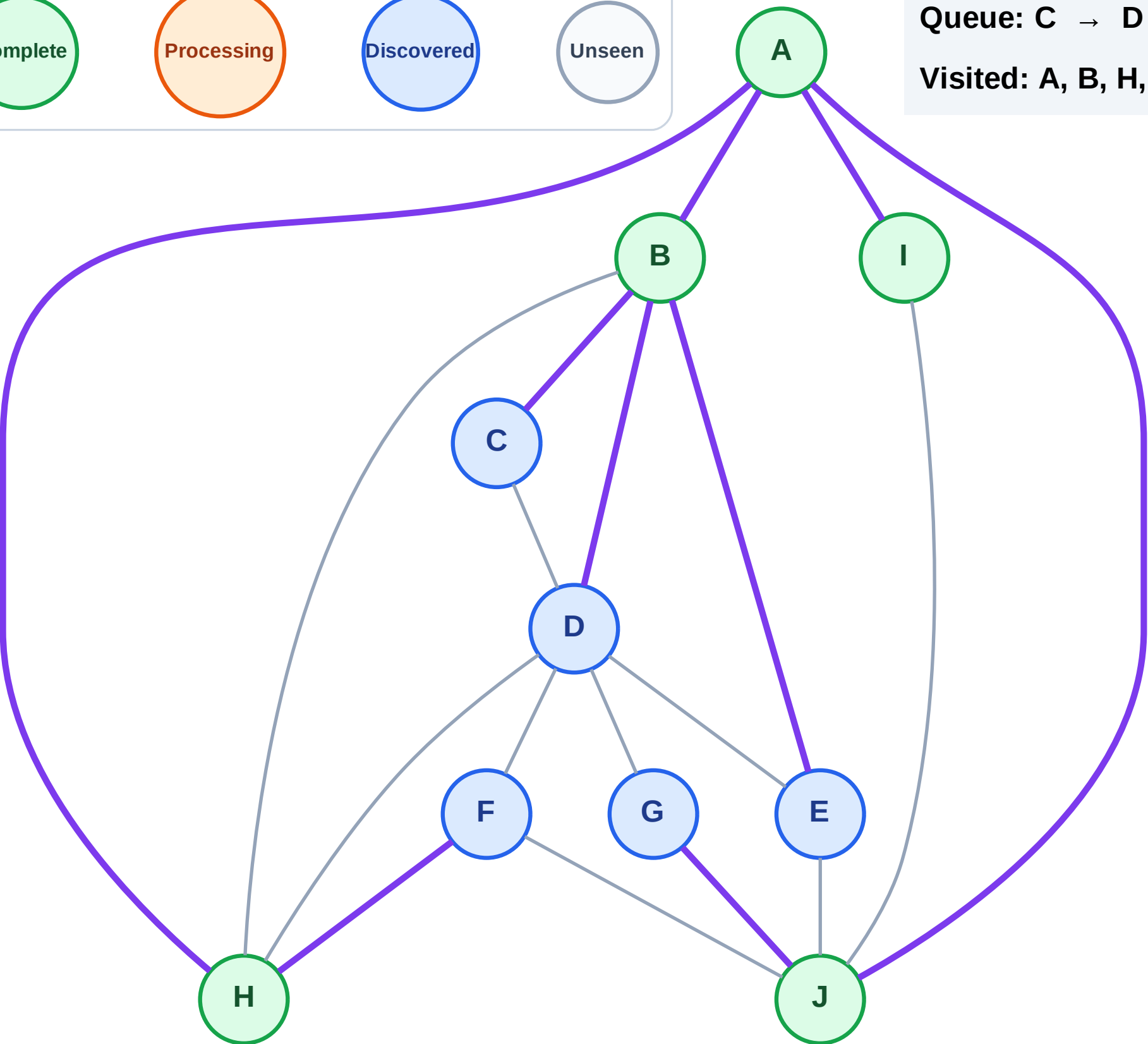
Processing

Discovered

Unseen

Queue: C → D → E → F → G

Visited: A, B, H, I, J



BFS Traversal

Step 12: Process node C

Legend

Complete

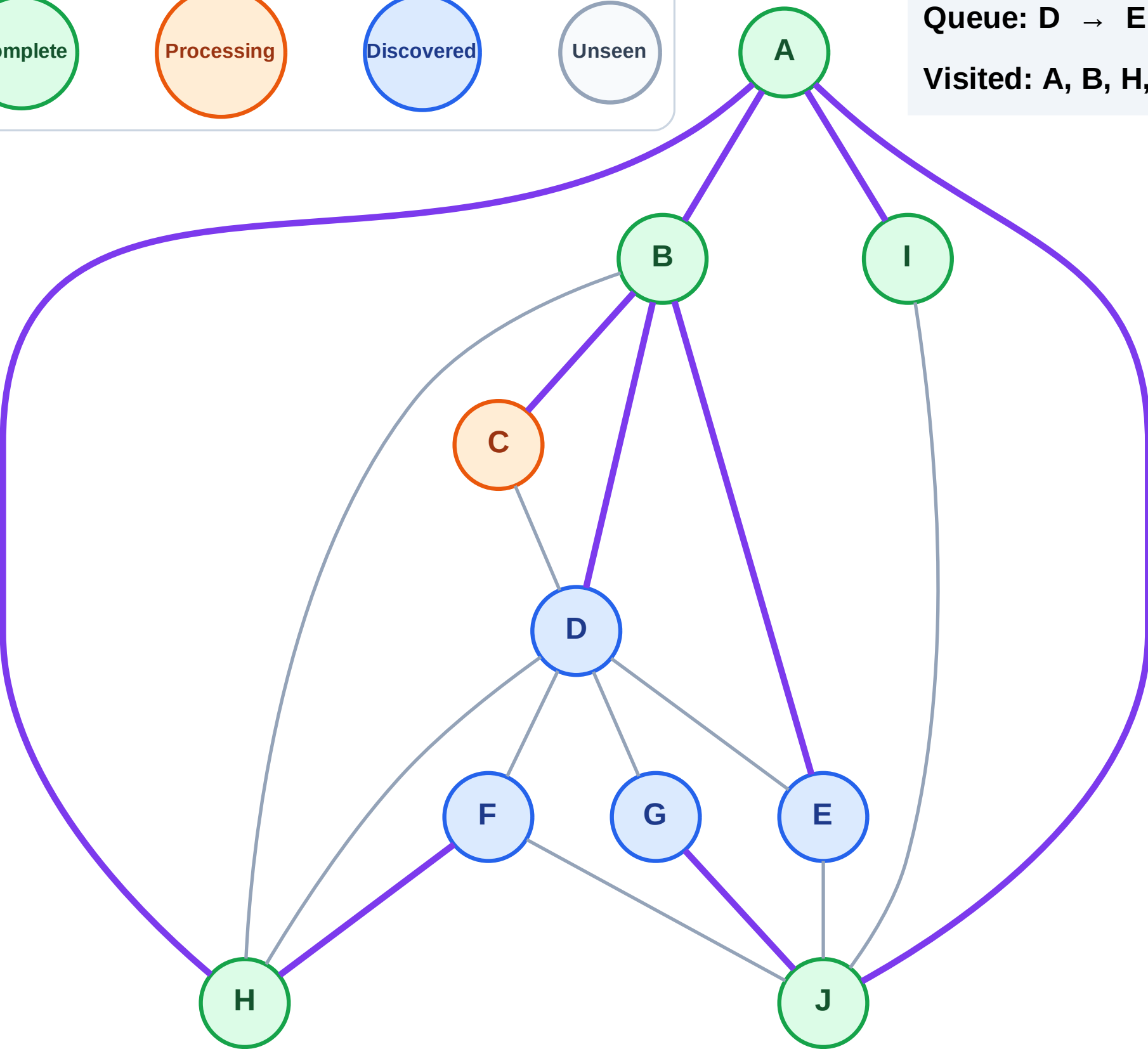
Processing

Discovered

Unseen

Queue: D → E → F → G

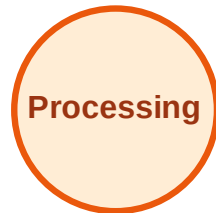
Visited: A, B, H, I, J



BFS Traversal

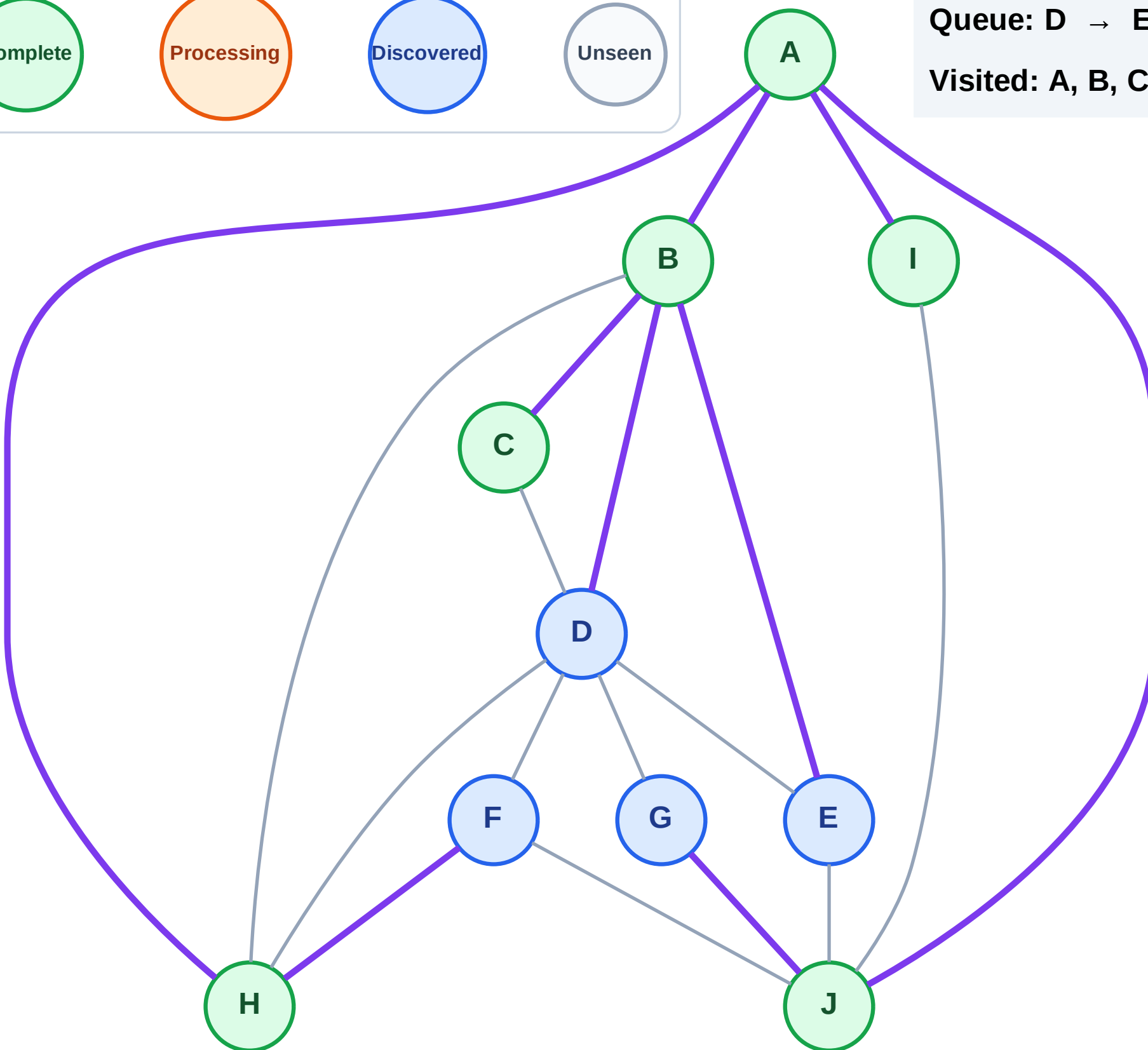
Step 13: No new neighbors from C

Legend



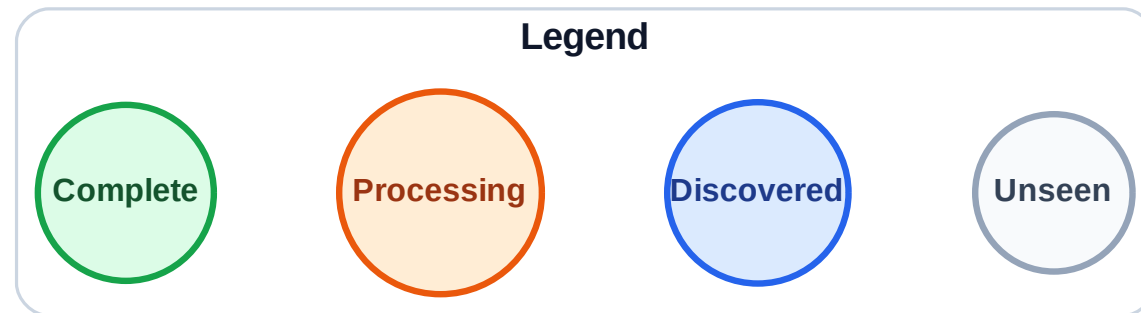
Queue: D → E → F → G

Visited: A, B, C, H, I, J

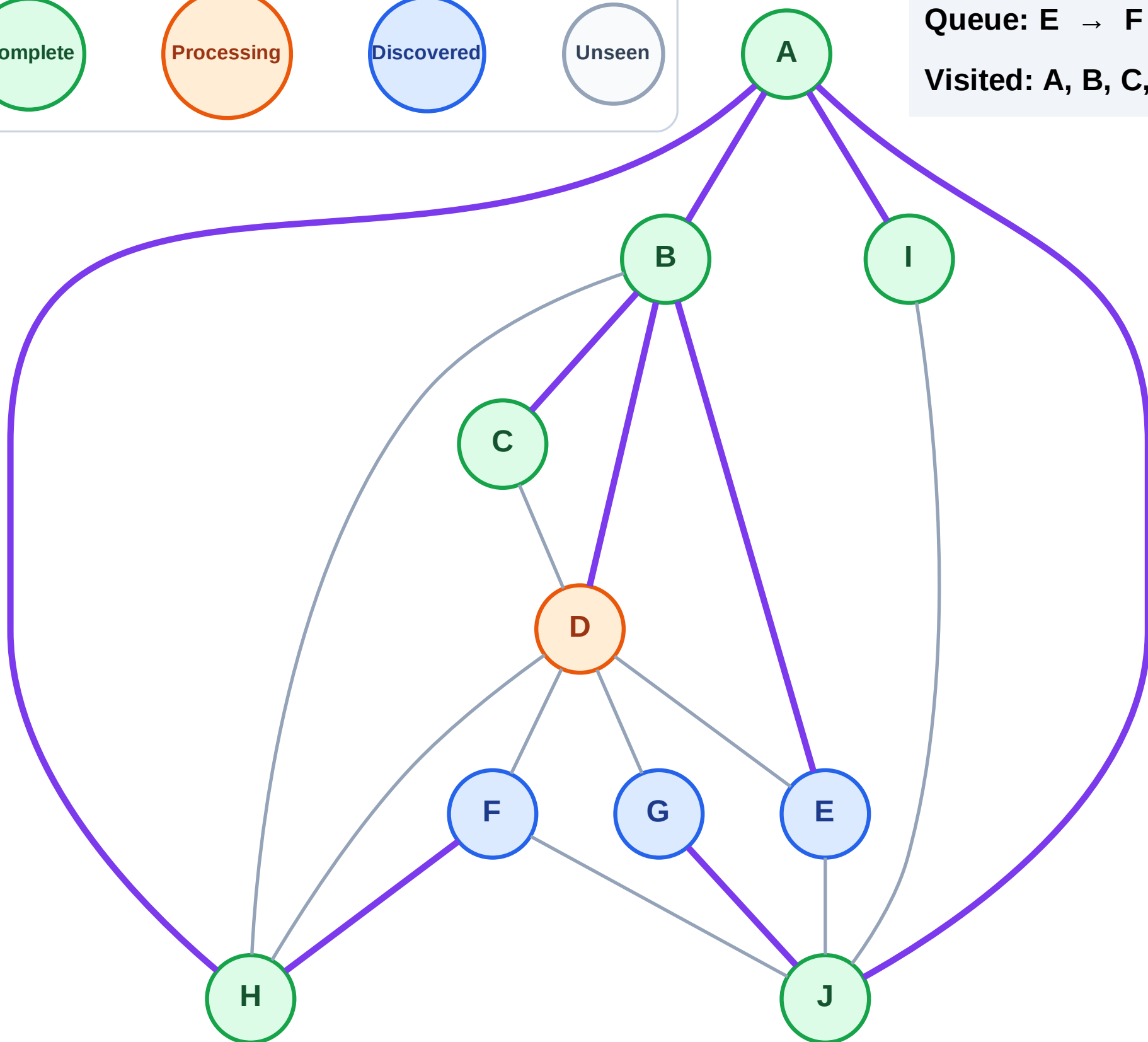


BFS Traversal

Step 14: Process node D



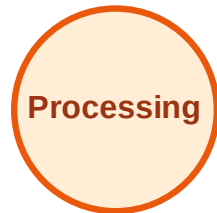
Queue: E → F → G
Visited: A, B, C, H, I, J



BFS Traversal

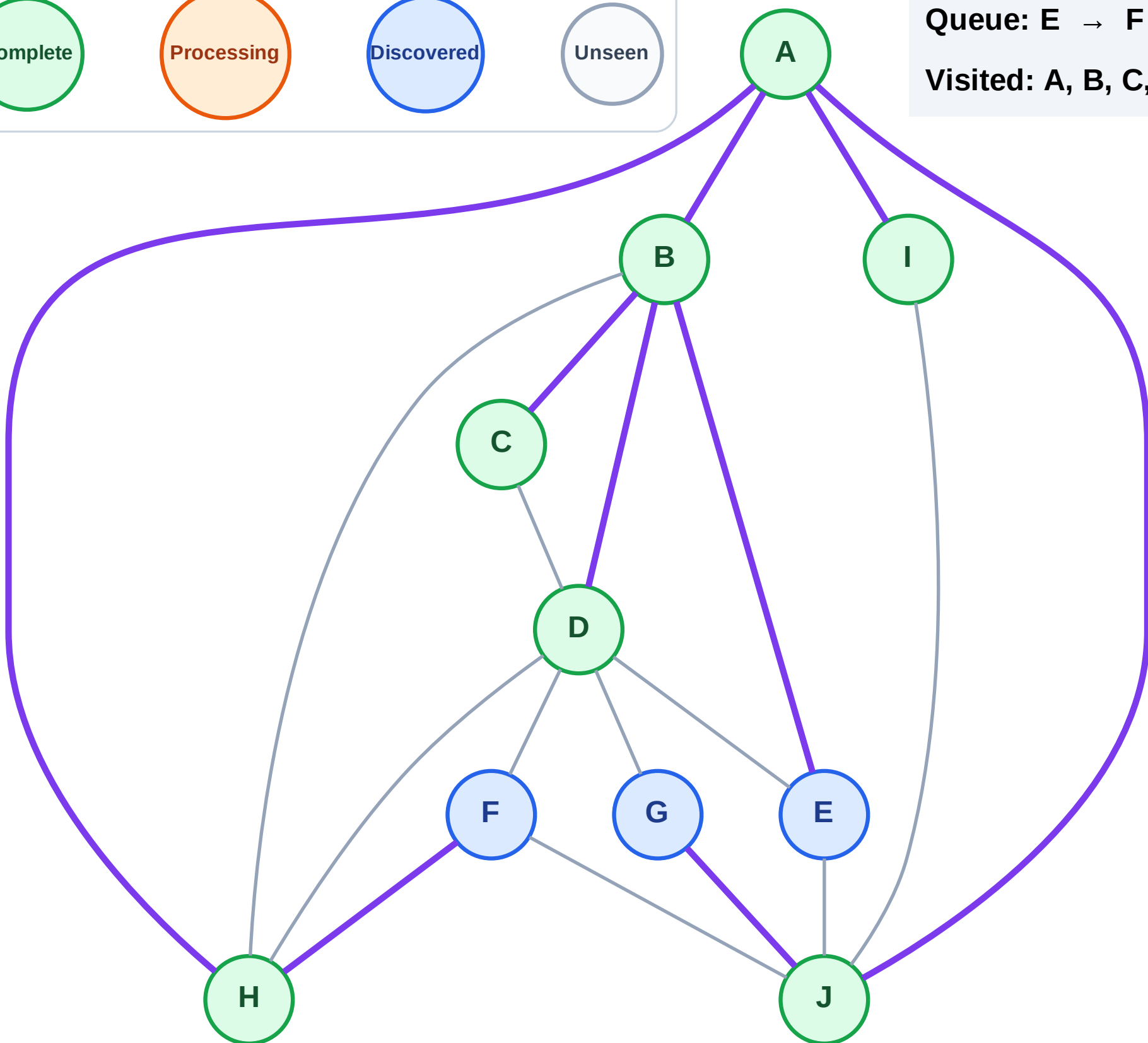
Step 15: No new neighbors from D

Legend



Queue: E → F → G

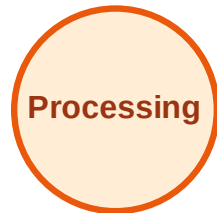
Visited: A, B, C, D, H, I, J



BFS Traversal

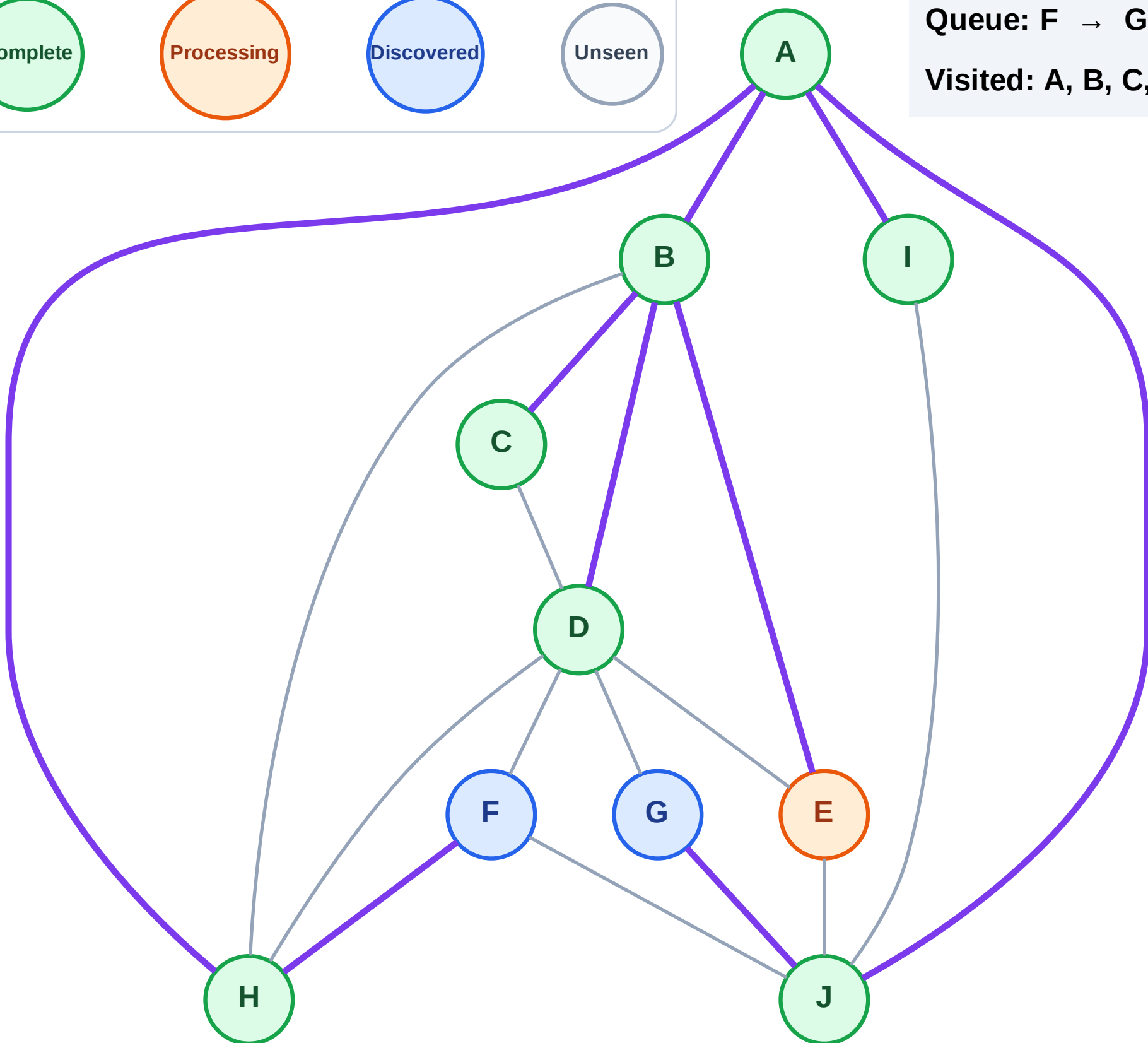
Step 16: Process node E

Legend



Queue: F → G

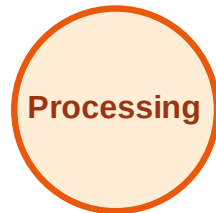
Visited: A, B, C, D, H, I, J



BFS Traversal

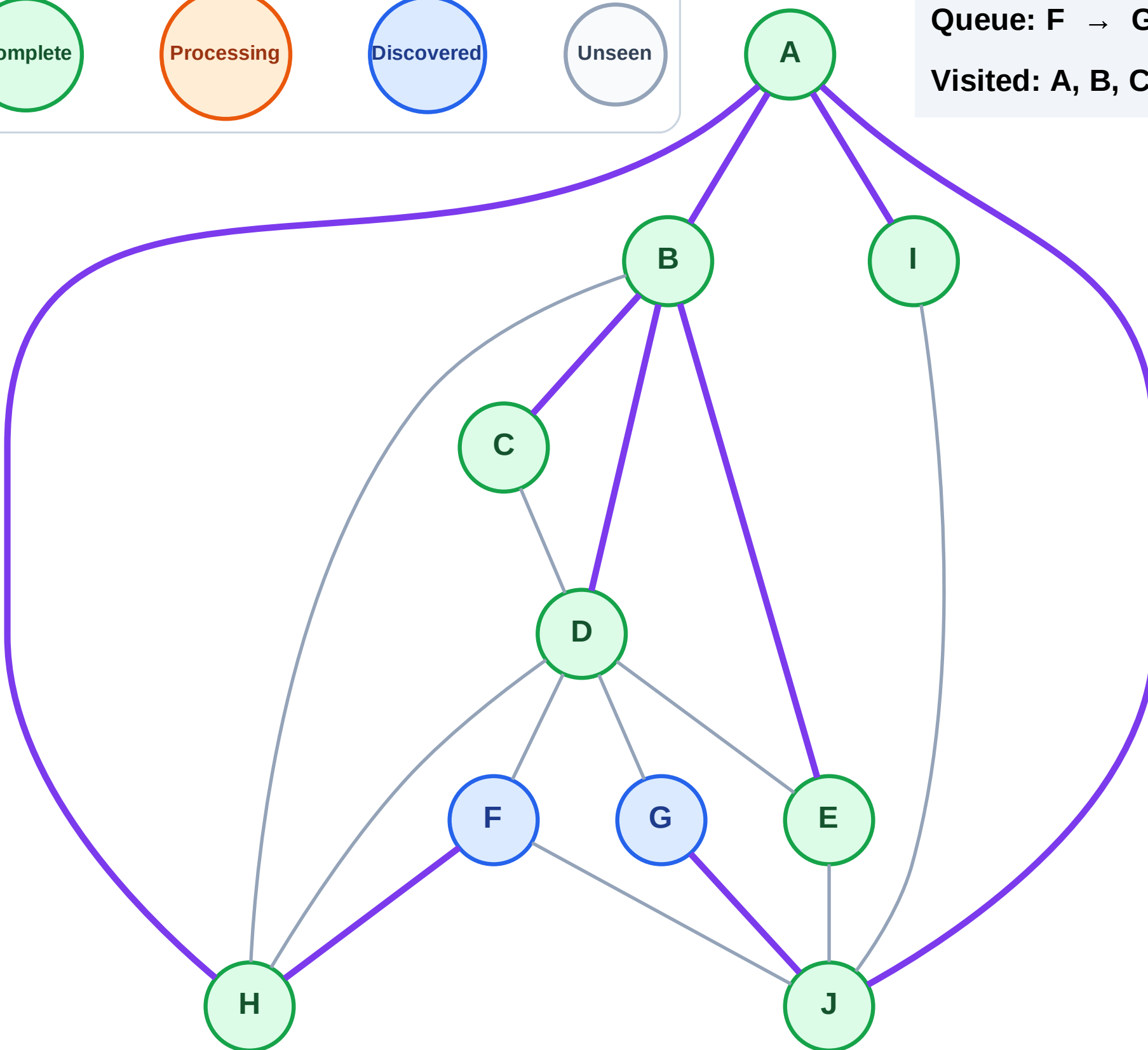
Step 17: No new neighbors from E

Legend



Queue: F → G

Visited: A, B, C, D, E, H, I, J



BFS Traversal

Step 18: Process node F

Legend

Complete

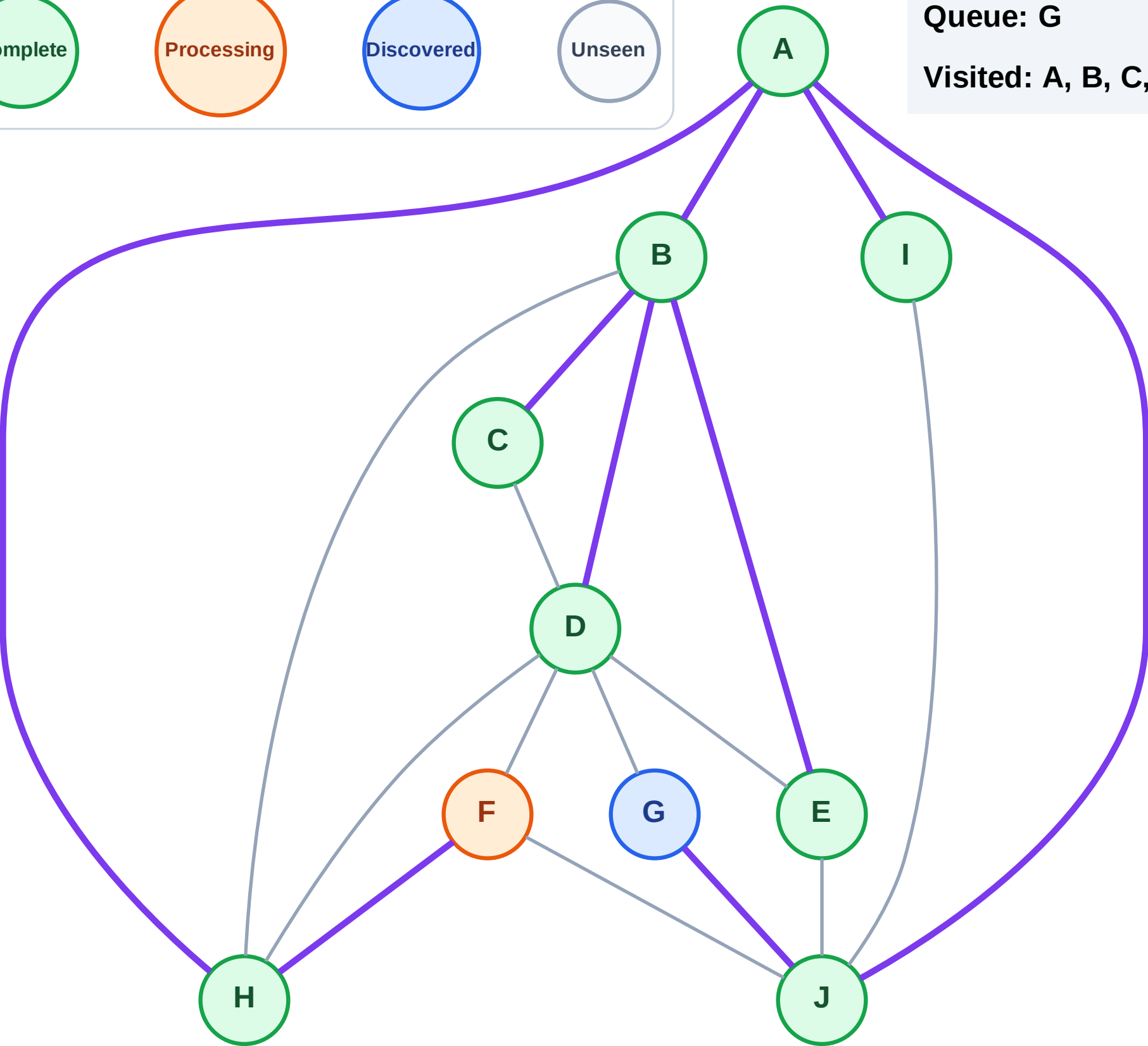
Processing

Discovered

Unseen

Queue: G

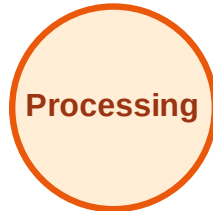
Visited: A, B, C, D, E, H, I, J



BFS Traversal

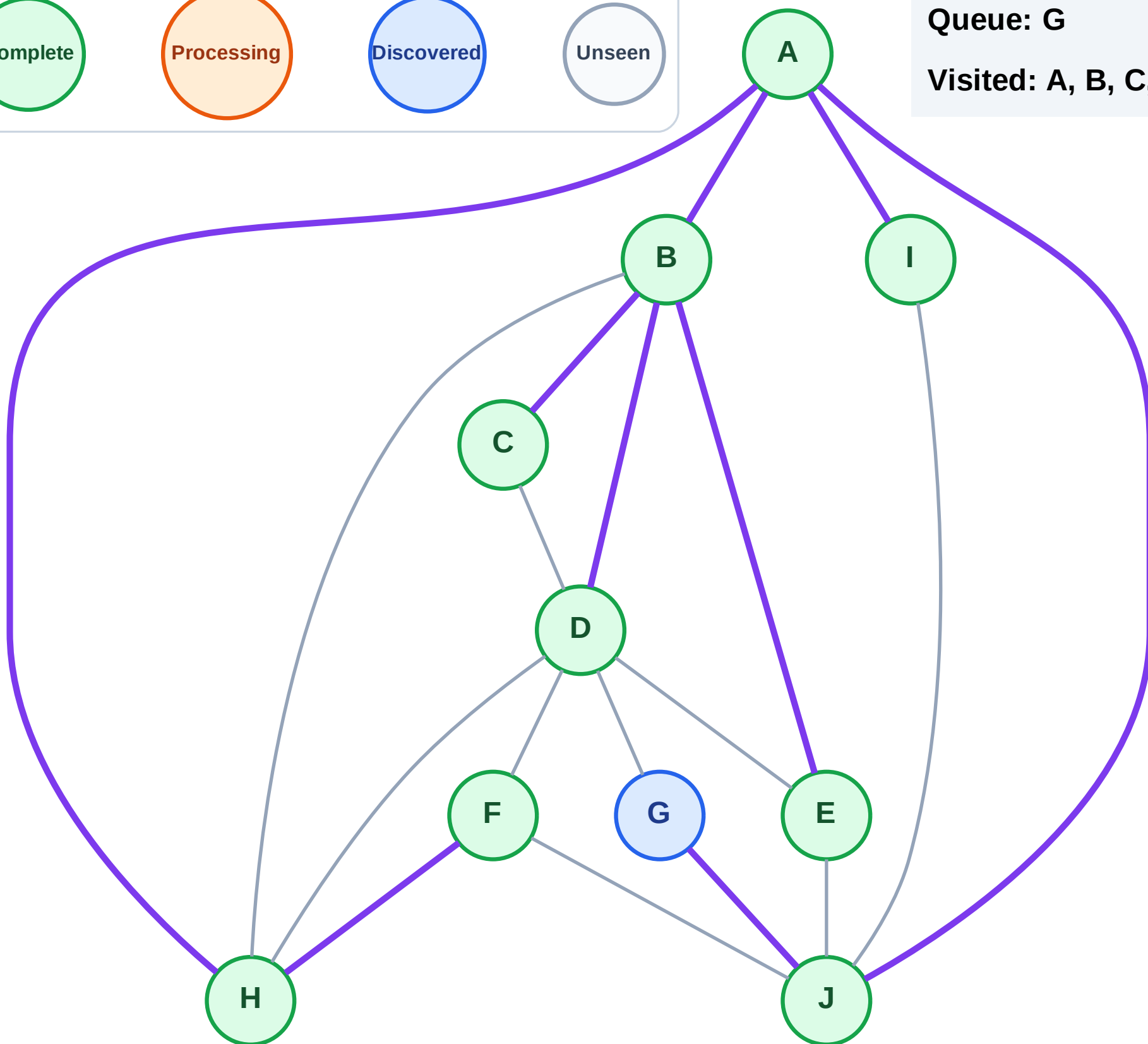
Step 19: No new neighbors from F

Legend



Queue: G

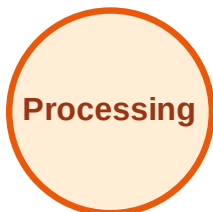
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

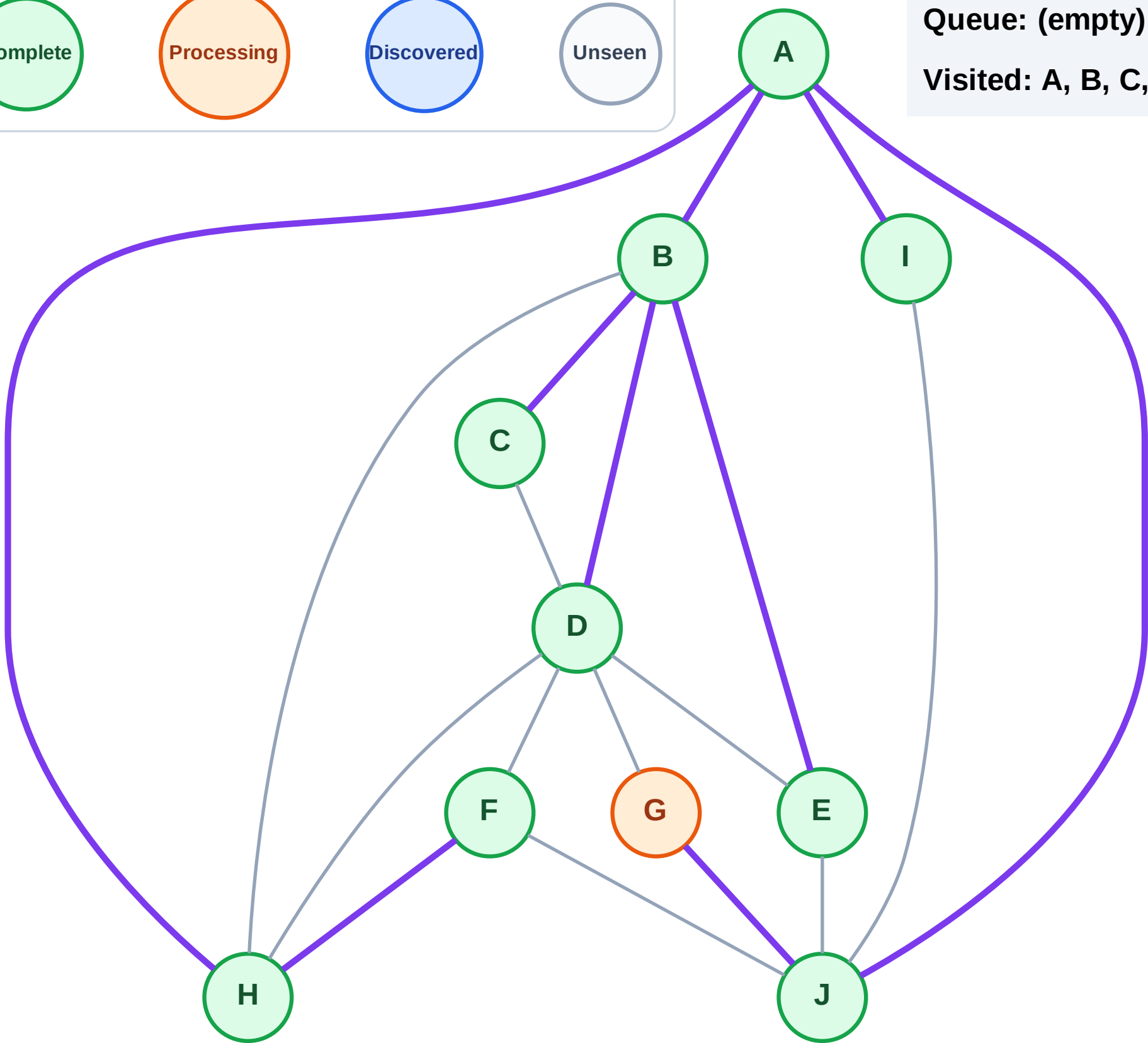
Step 20: Process node G

Legend



Queue: (empty)

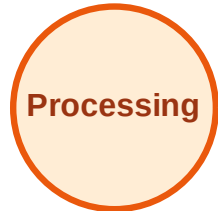
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

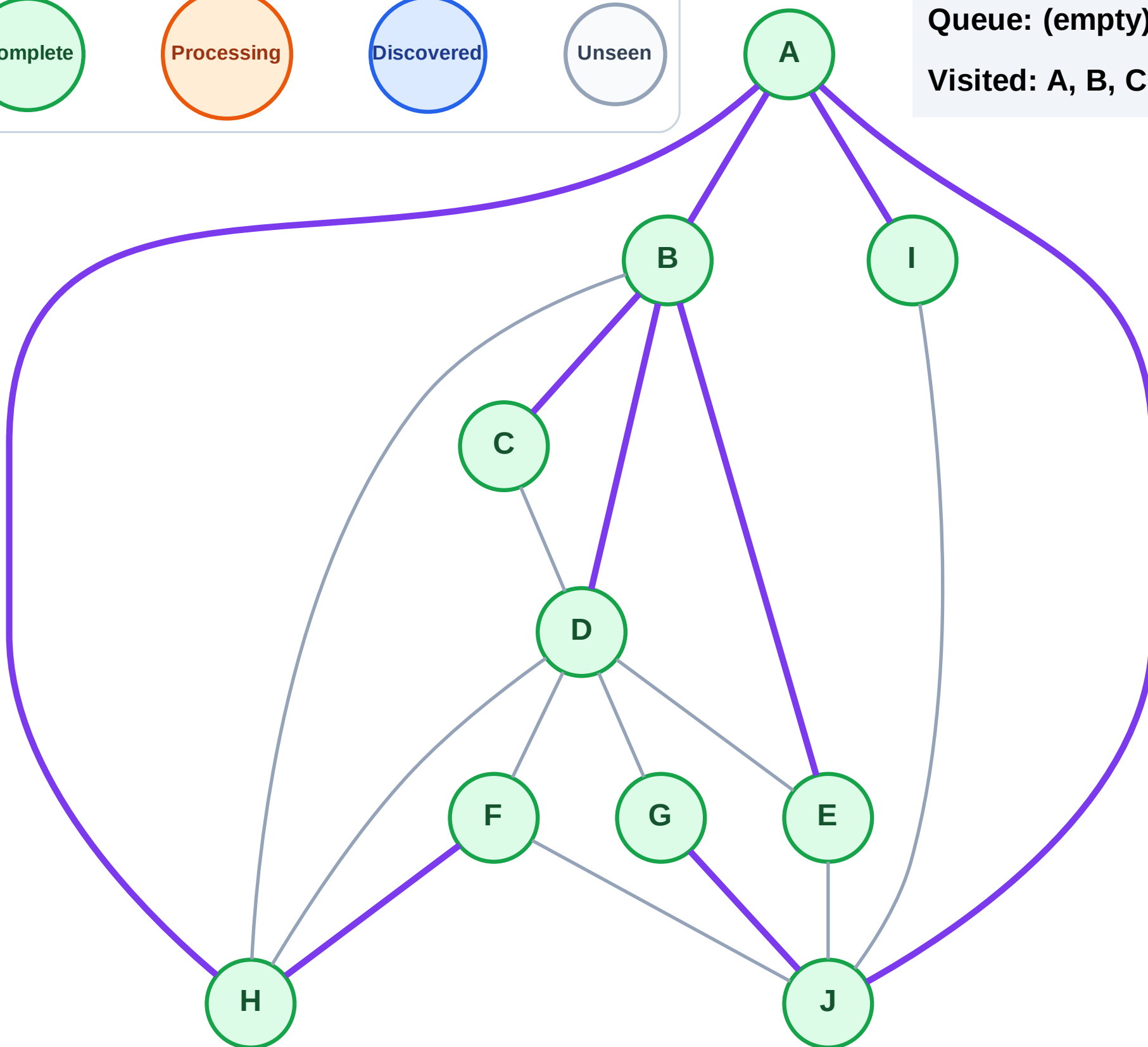
Step 21: No new neighbors from G

Legend



Queue: (empty)

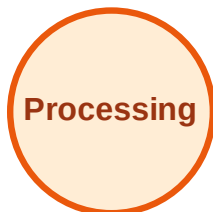
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

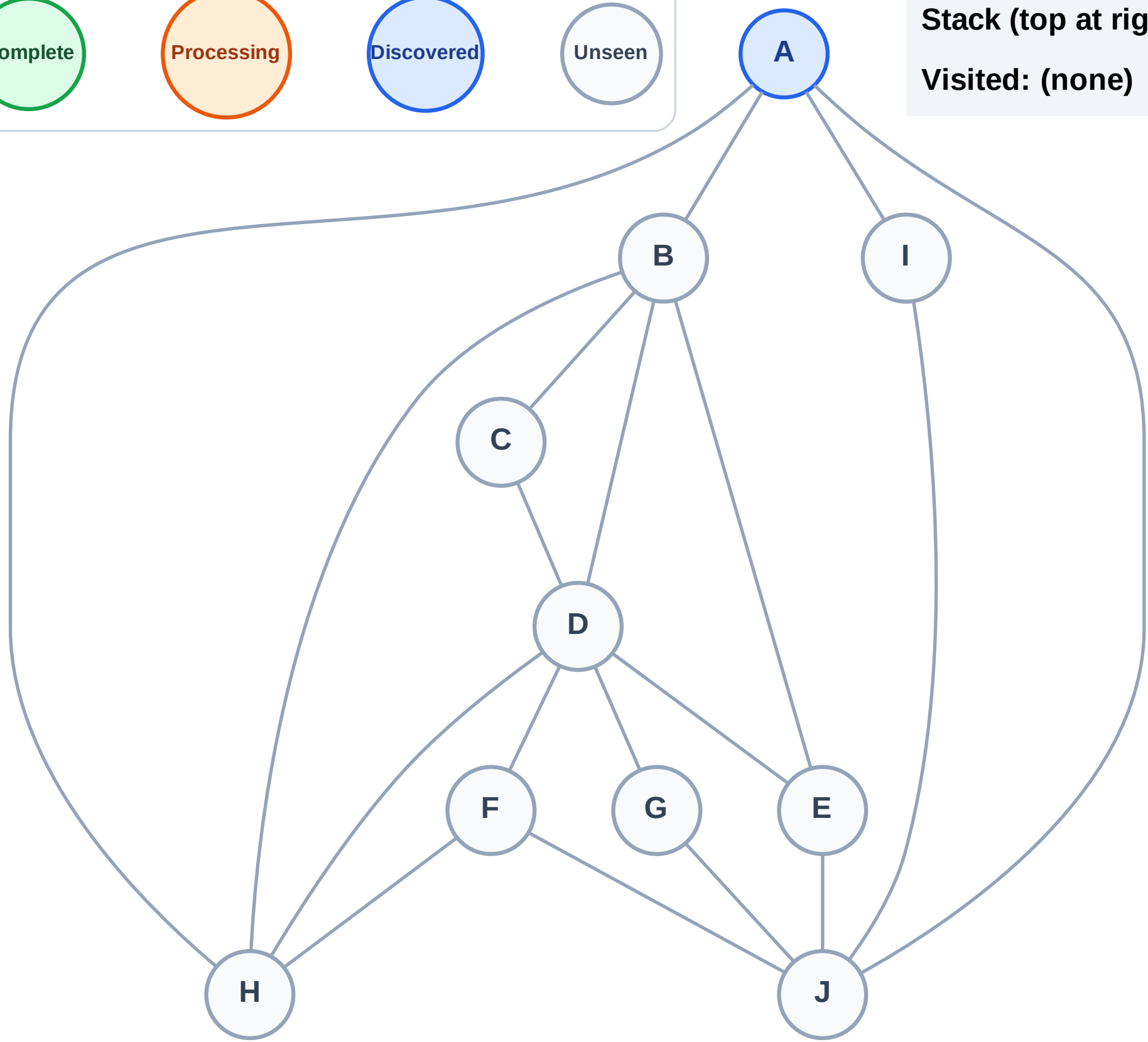
Step 1: Start at A

Legend



Stack (top at right): A

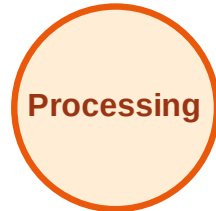
Visited: (none)



DFS Traversal

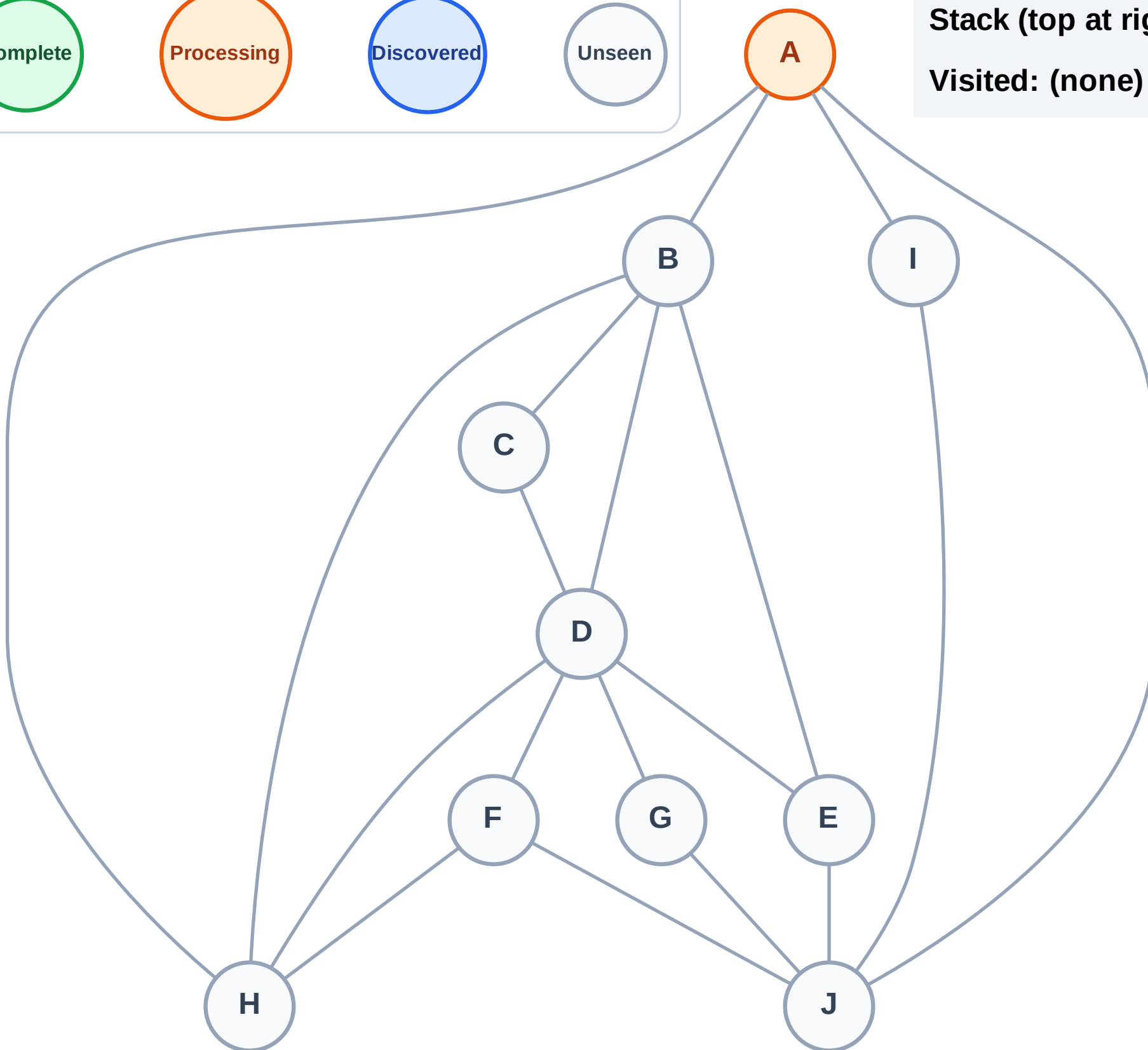
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover J, I, H, B from A

Legend

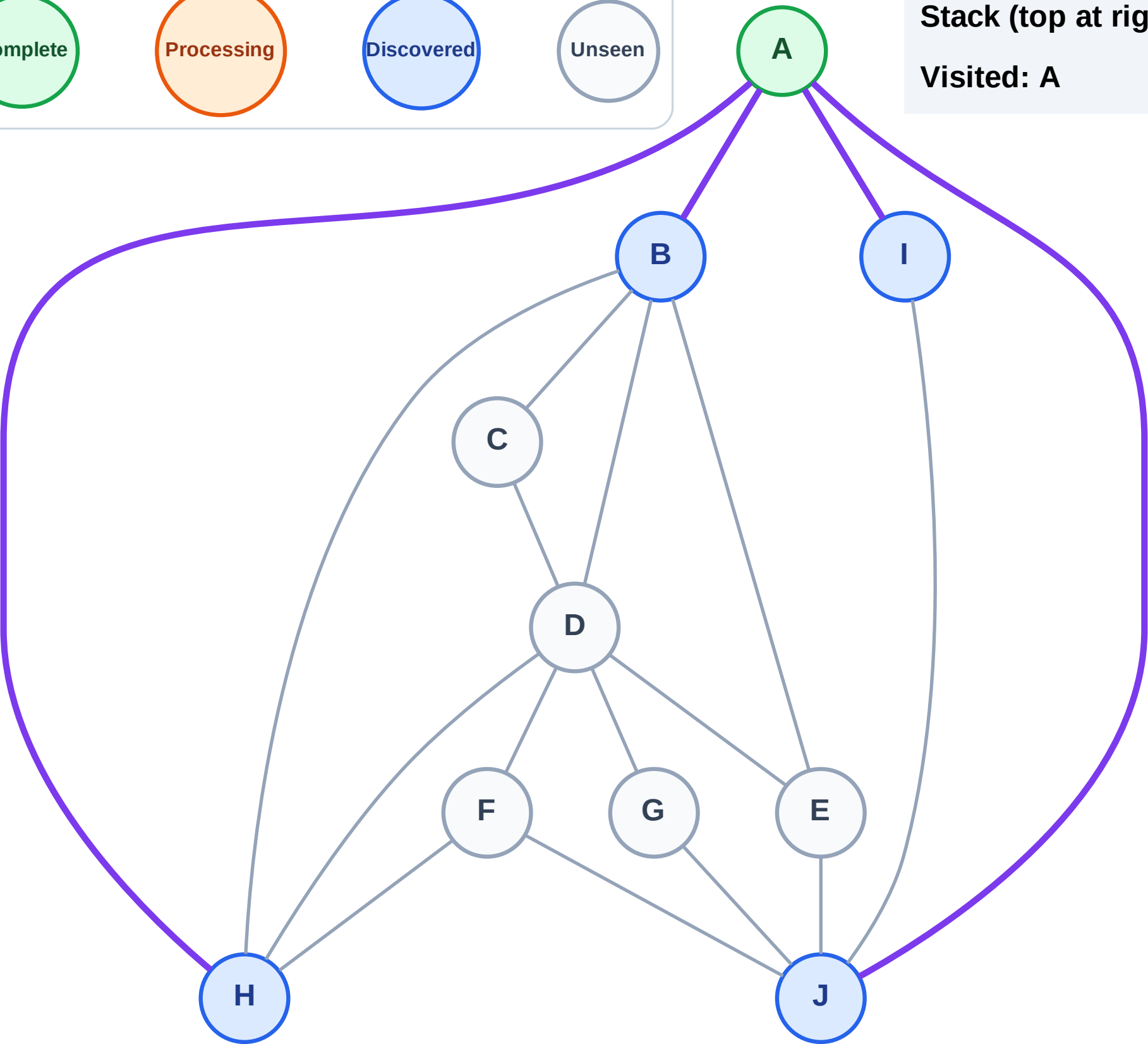
Complete

Processing

Discovered

Unseen

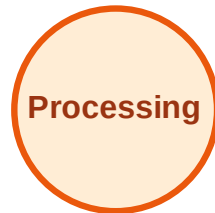
Stack (top at right): J | I | H | B
Visited: A



DFS Traversal

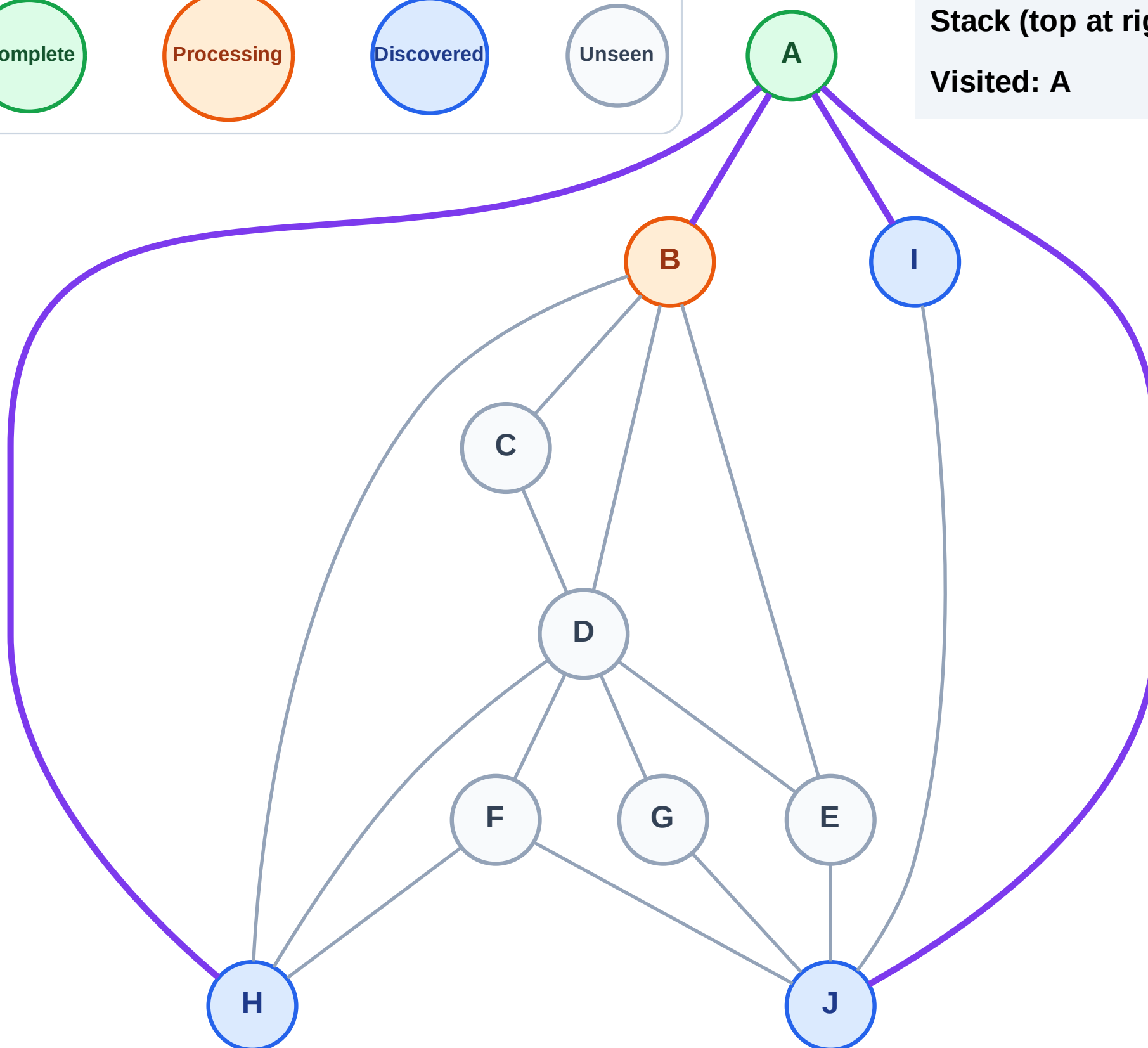
Step 4: Process node B

Legend



Stack (top at right): J | I | H

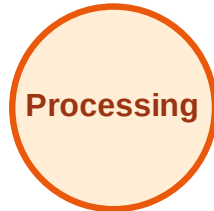
Visited: A



DFS Traversal

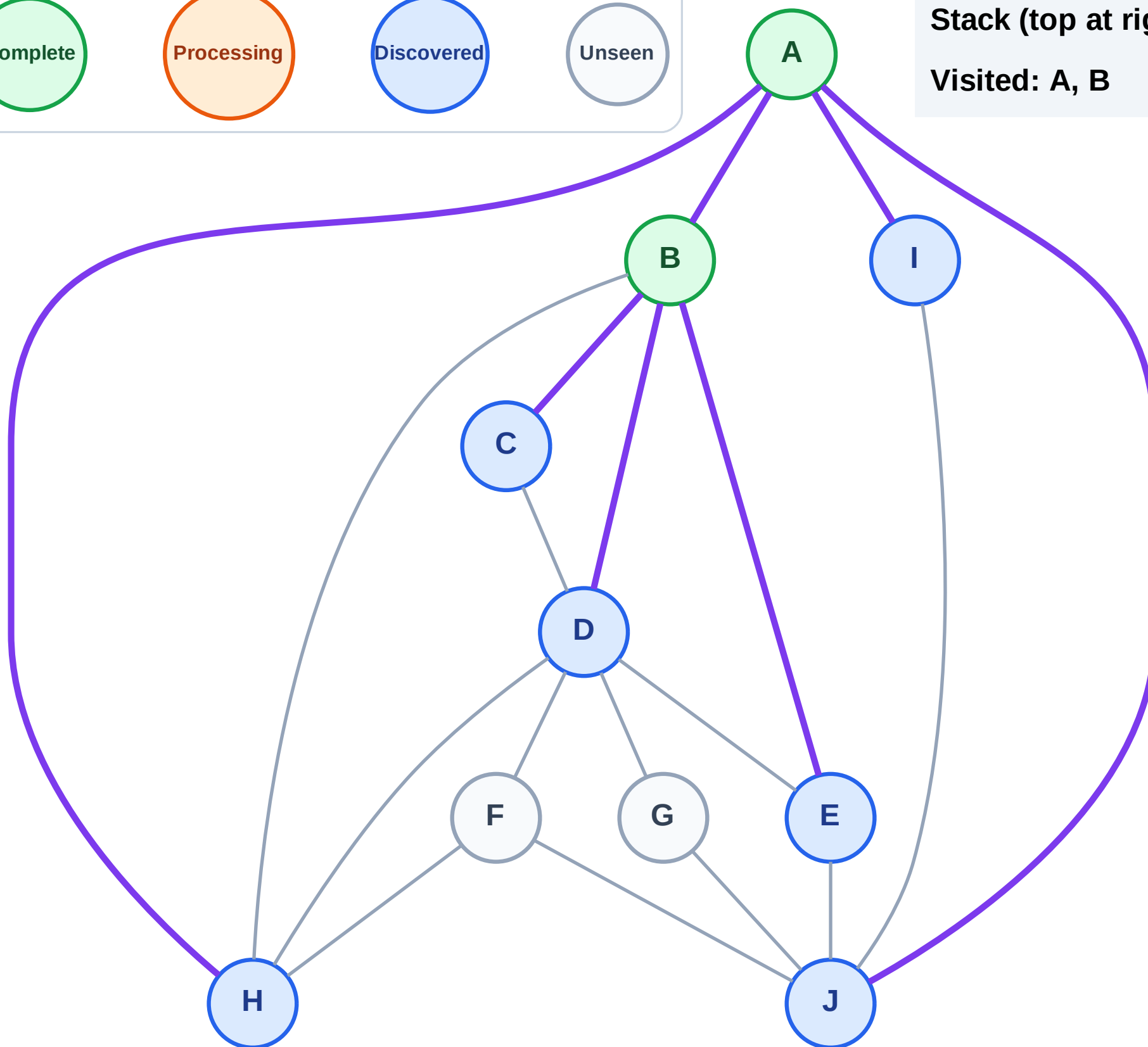
Step 5: Discover E, D, C from B

Legend



Stack (top at right): J | I | H | E | D | C

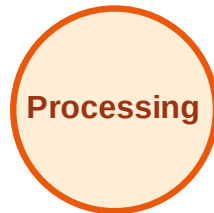
Visited: A, B



DFS Traversal

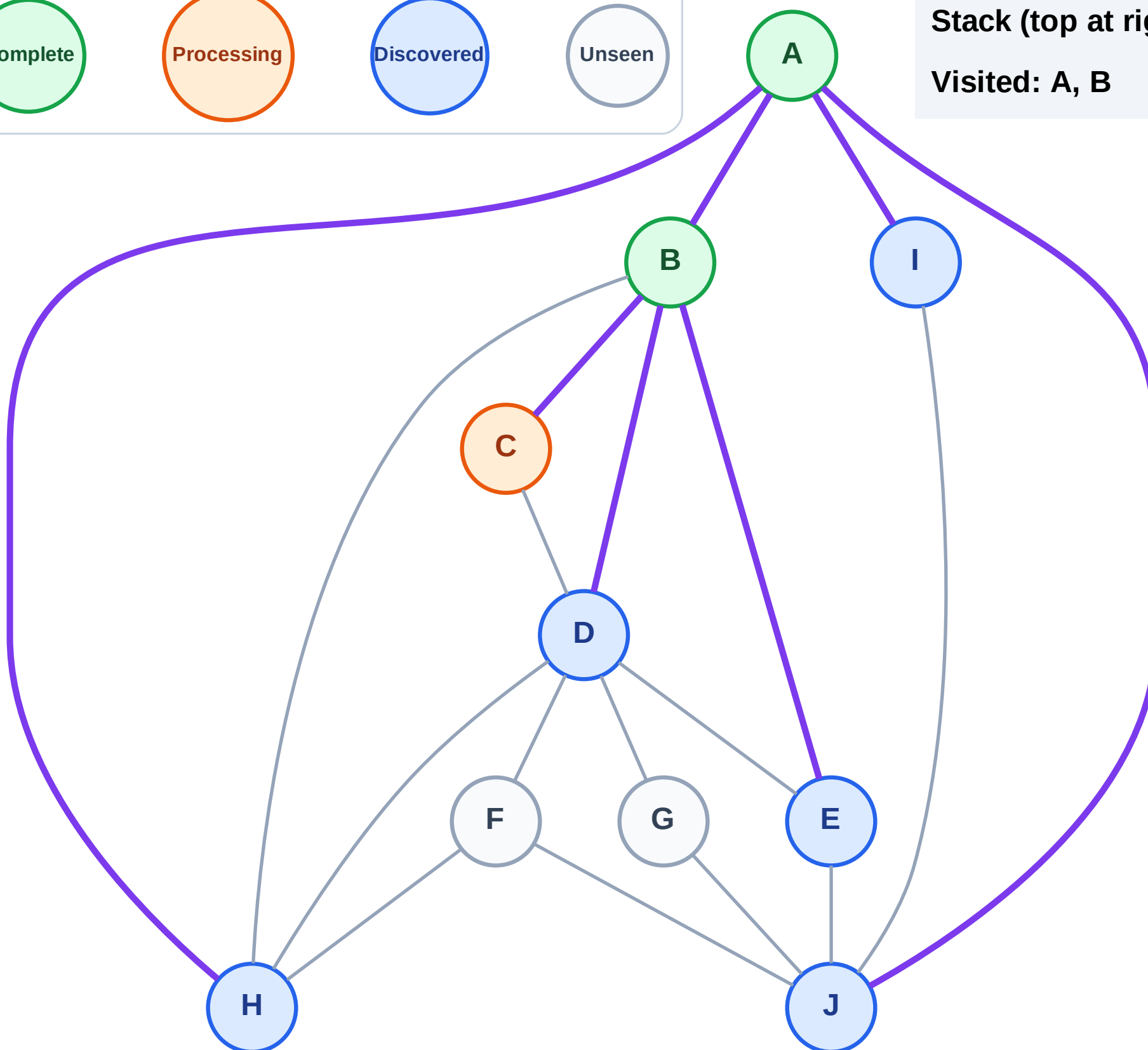
Step 6: Process node C

Legend



Stack (top at right): J | I | H | E | D

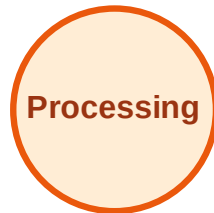
Visited: A, B



DFS Traversal

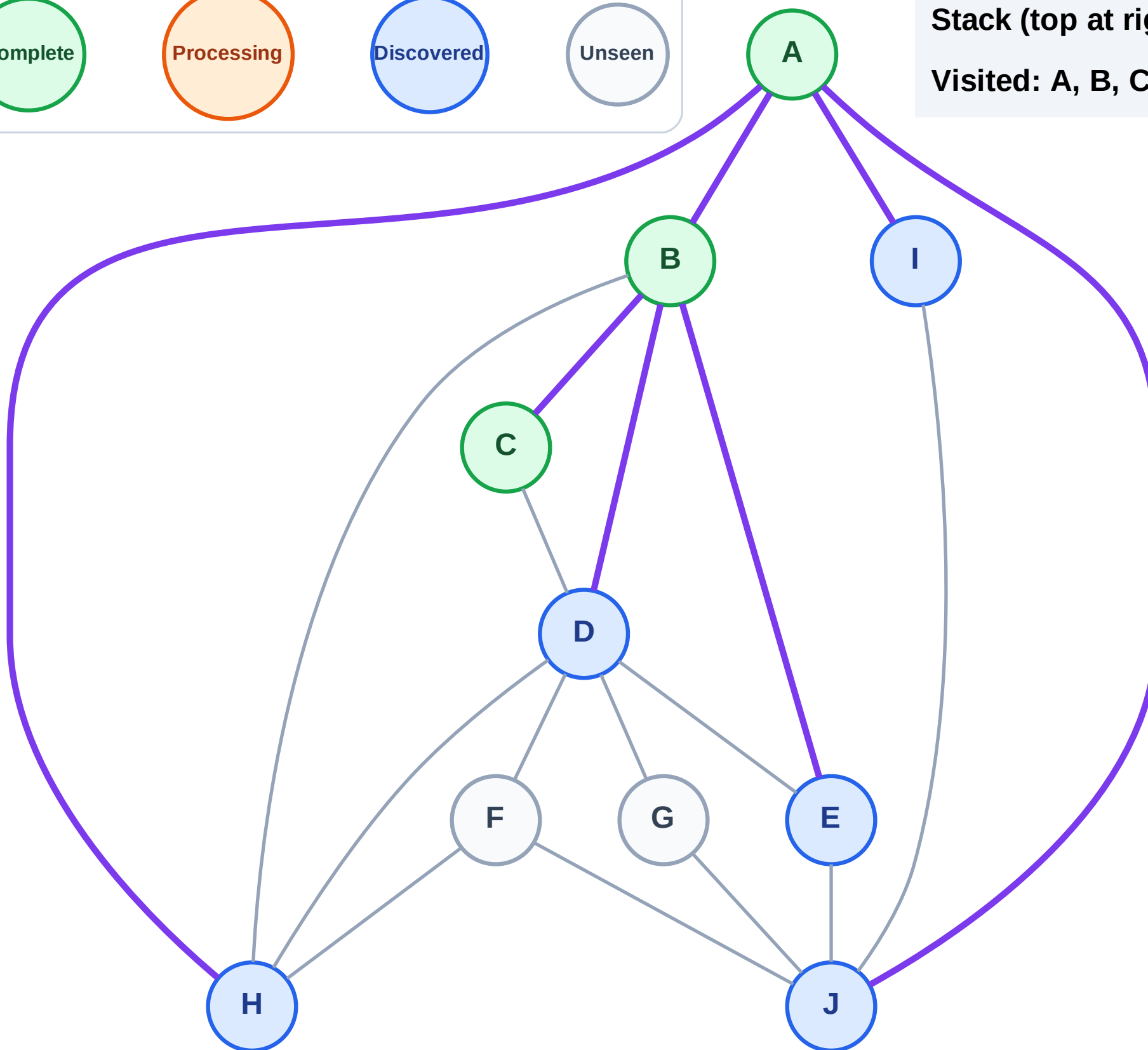
Step 7: No new neighbors from C

Legend



Stack (top at right): J | I | H | E | D

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

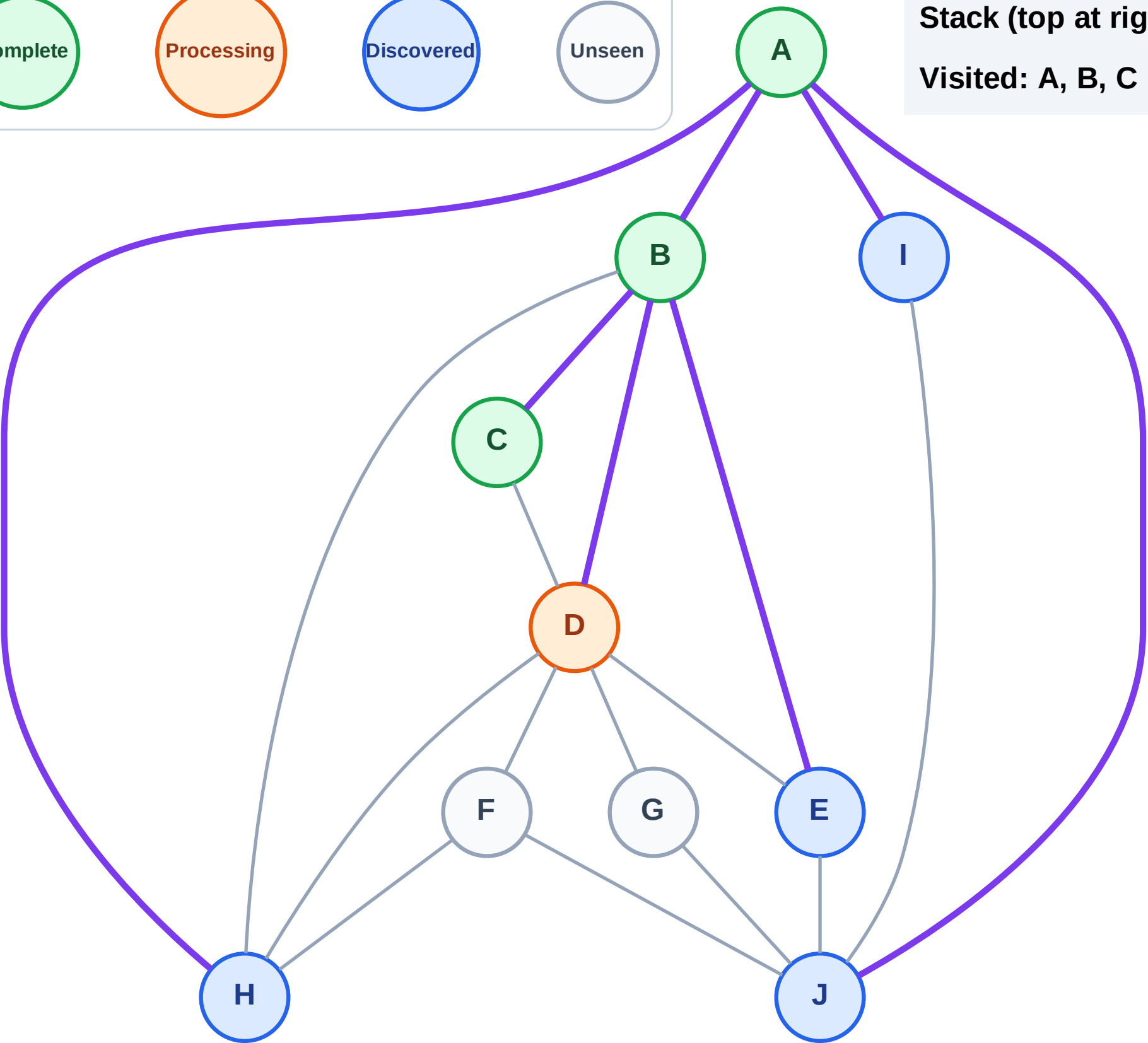
Processing

Discovered

Unseen

Stack (top at right): J | I | H | E

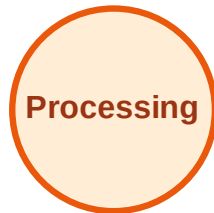
Visited: A, B, C



DFS Traversal

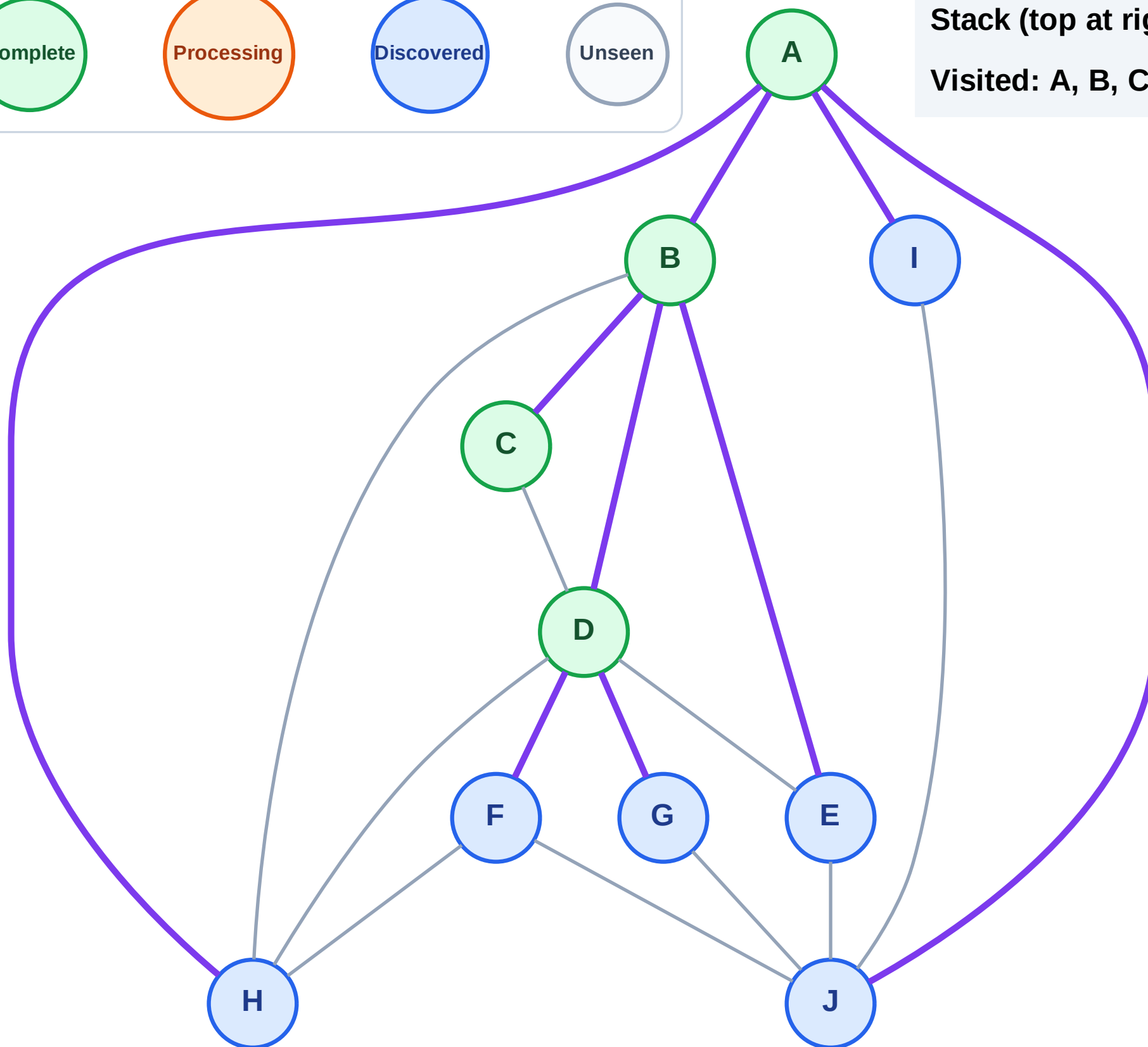
Step 9: Discover G, F from D

Legend



Stack (top at right): J | I | H | E | G | F

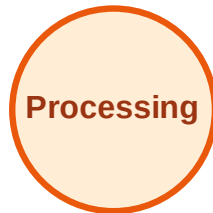
Visited: A, B, C, D



DFS Traversal

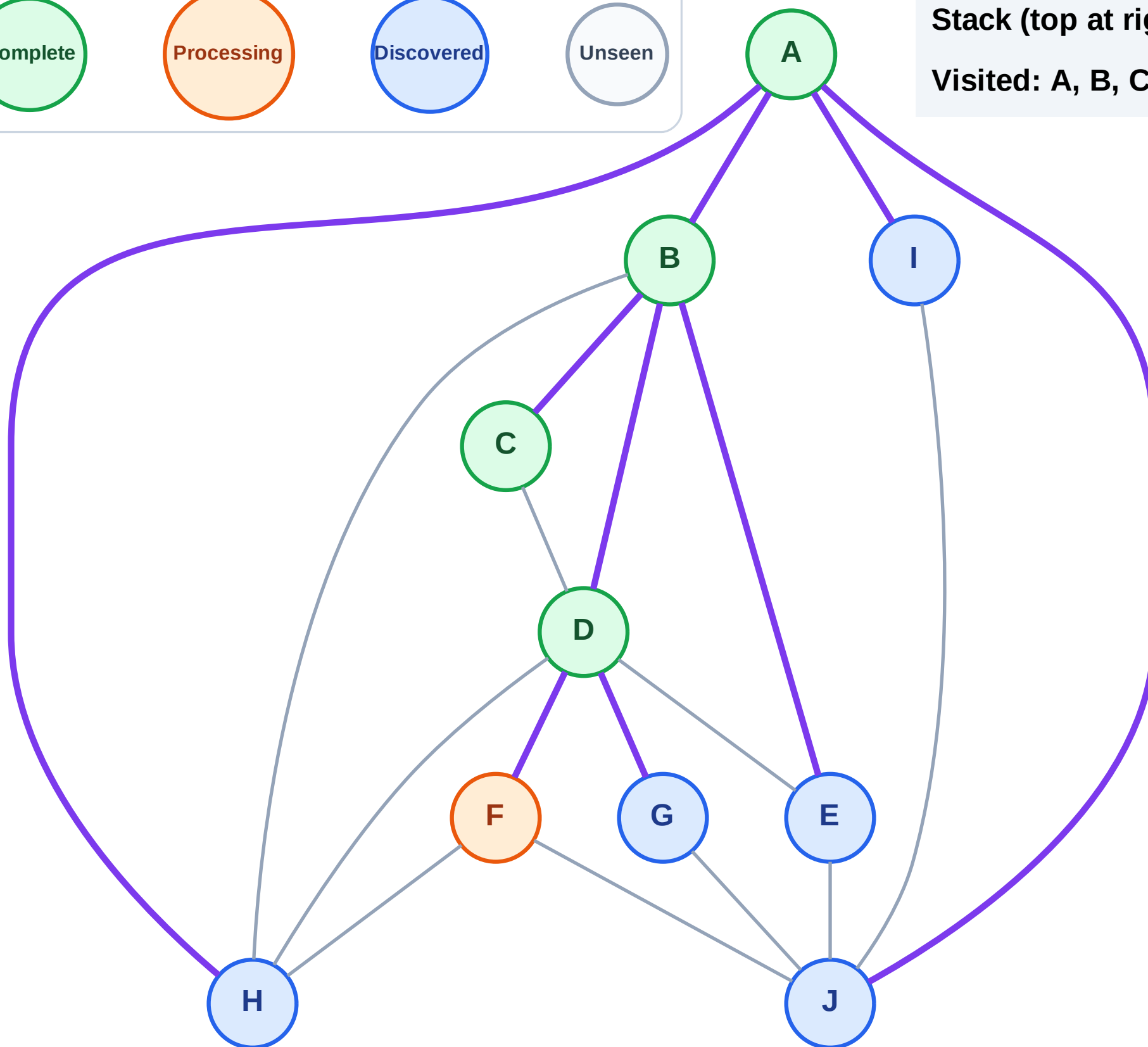
Step 10: Process node F

Legend



Stack (top at right): J | I | H | E | G

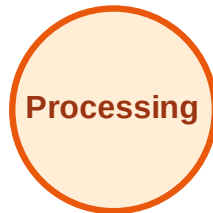
Visited: A, B, C, D



DFS Traversal

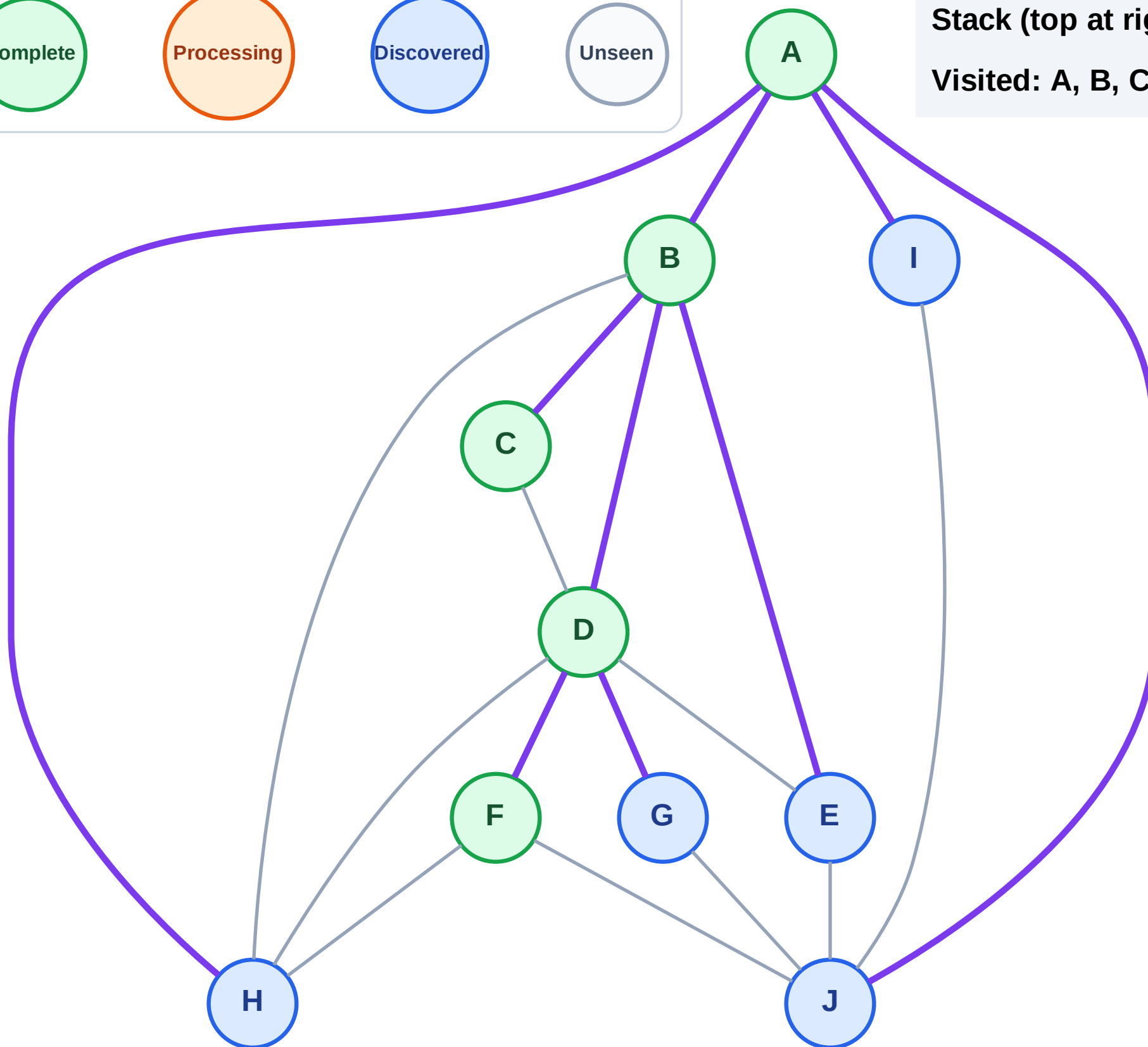
Step 11: No new neighbors from F

Legend



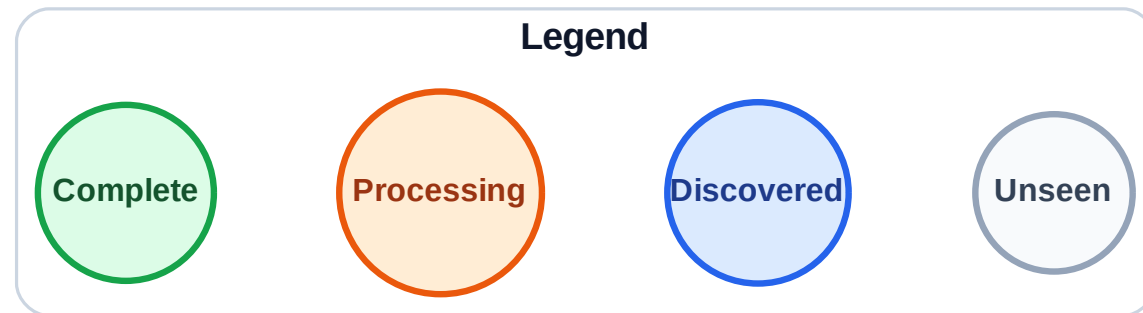
Stack (top at right): J | I | H | E | G

Visited: A, B, C, D, F

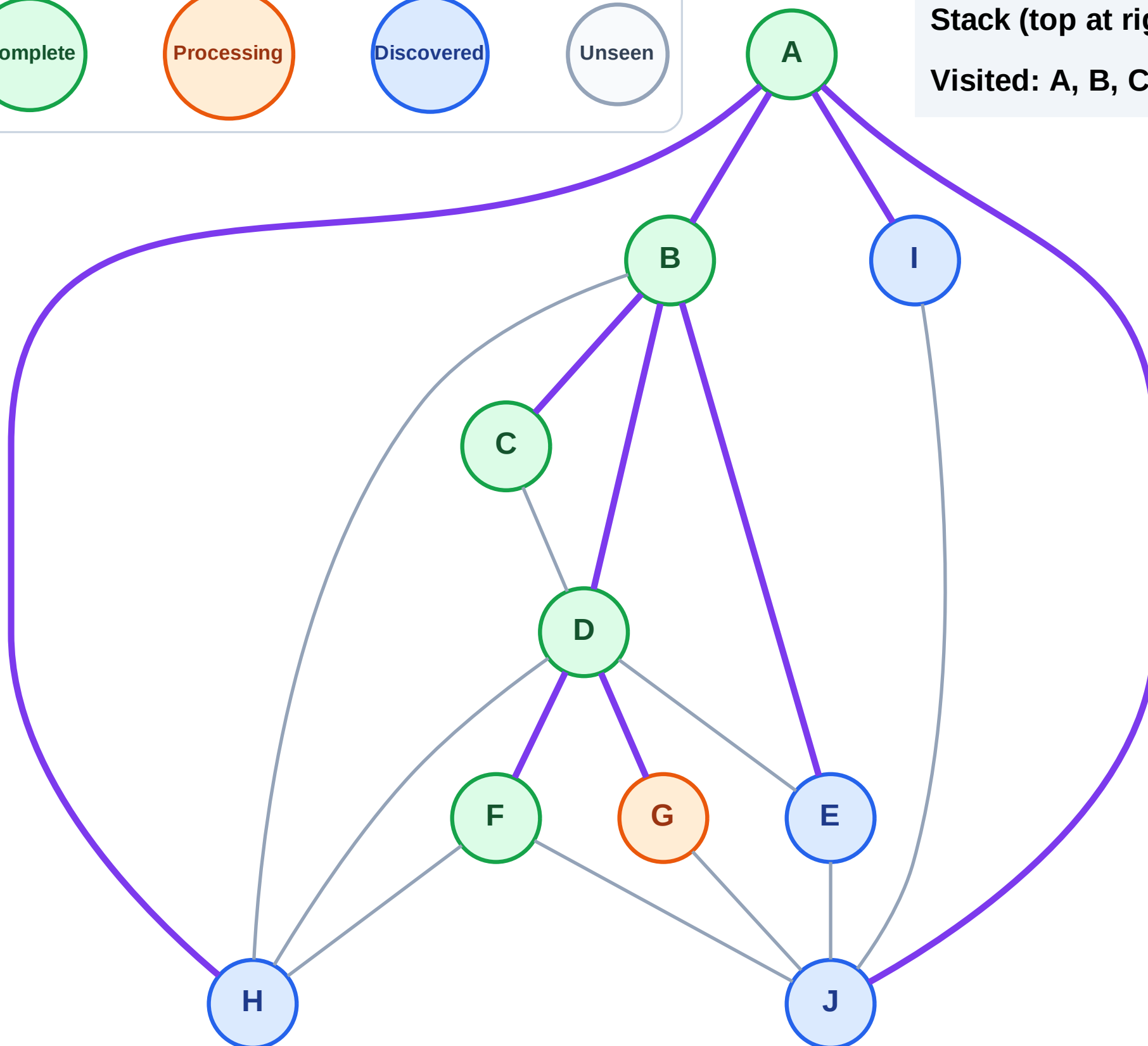


DFS Traversal

Step 12: Process node G



Stack (top at right): J | I | H | E
Visited: A, B, C, D, F



DFS Traversal

Step 13: No new neighbors from G

Legend

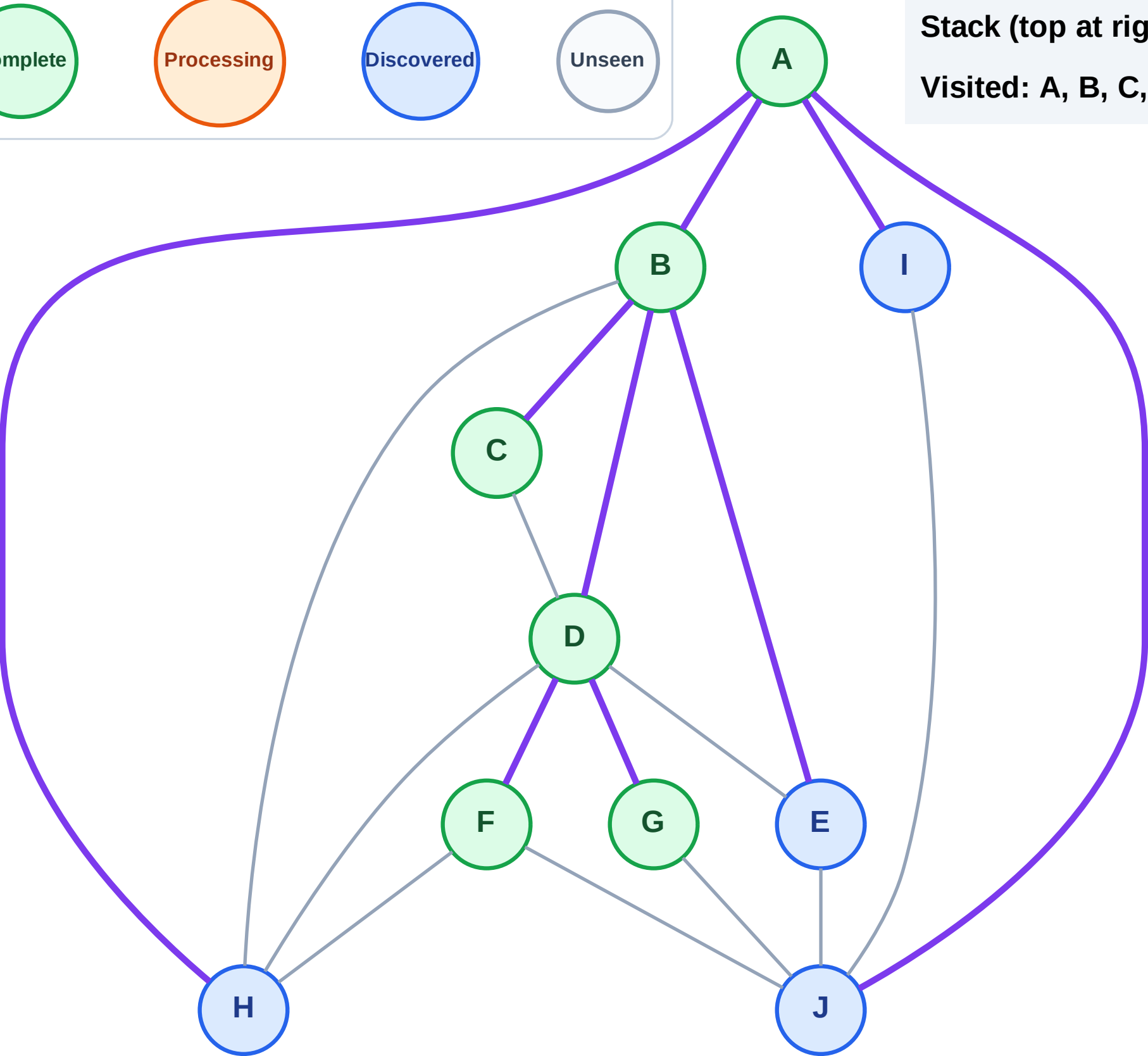
Complete

Processing

Discovered

Unseen

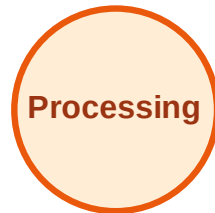
Stack (top at right): J | I | H | E
Visited: A, B, C, D, F, G



DFS Traversal

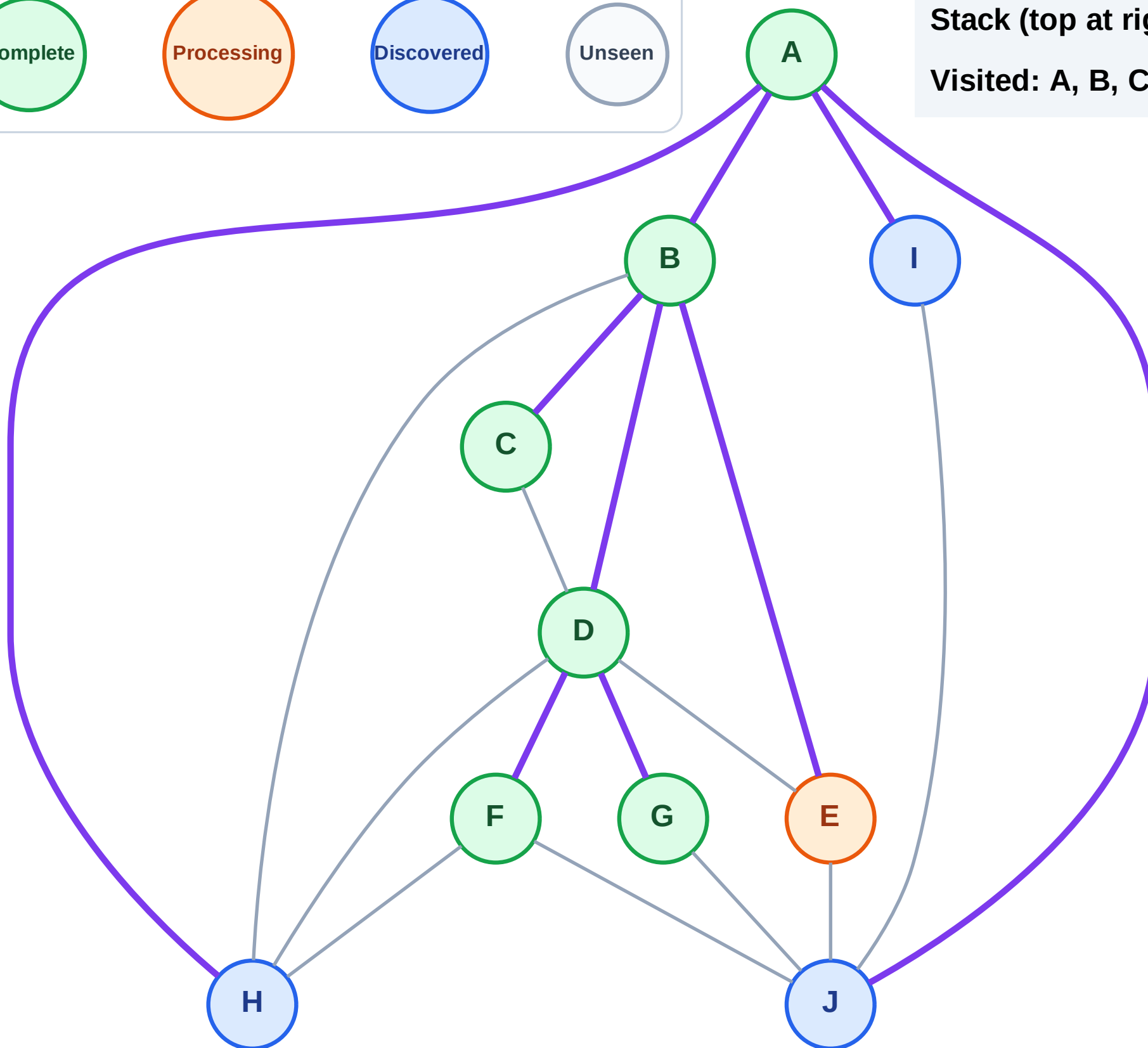
Step 14: Process node E

Legend



Stack (top at right): J | I | H

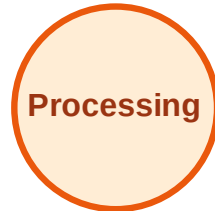
Visited: A, B, C, D, F, G



DFS Traversal

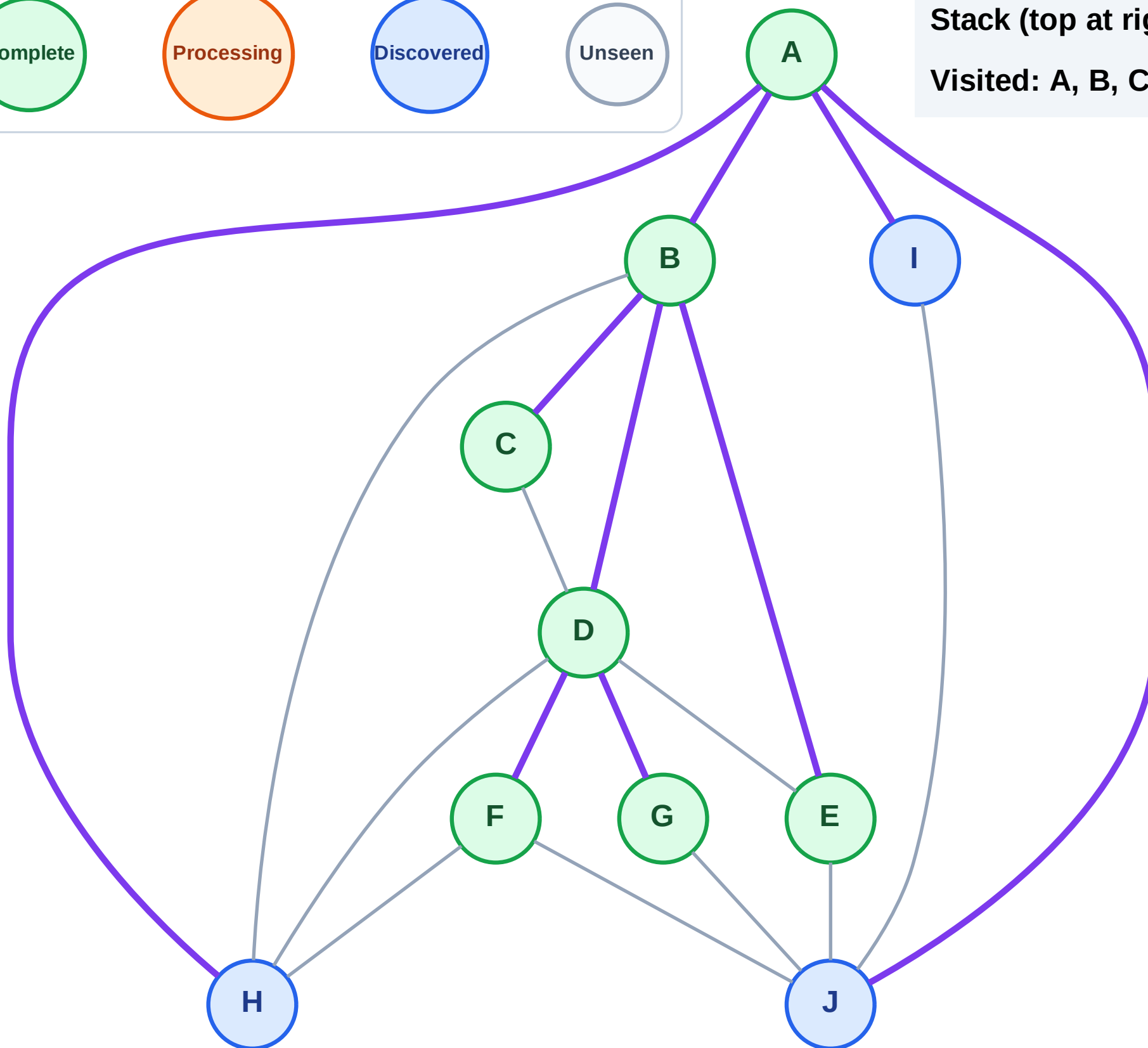
Step 15: No new neighbors from E

Legend



Stack (top at right): J | I | H

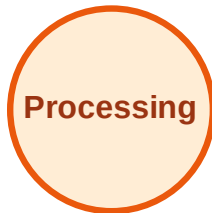
Visited: A, B, C, D, E, F, G



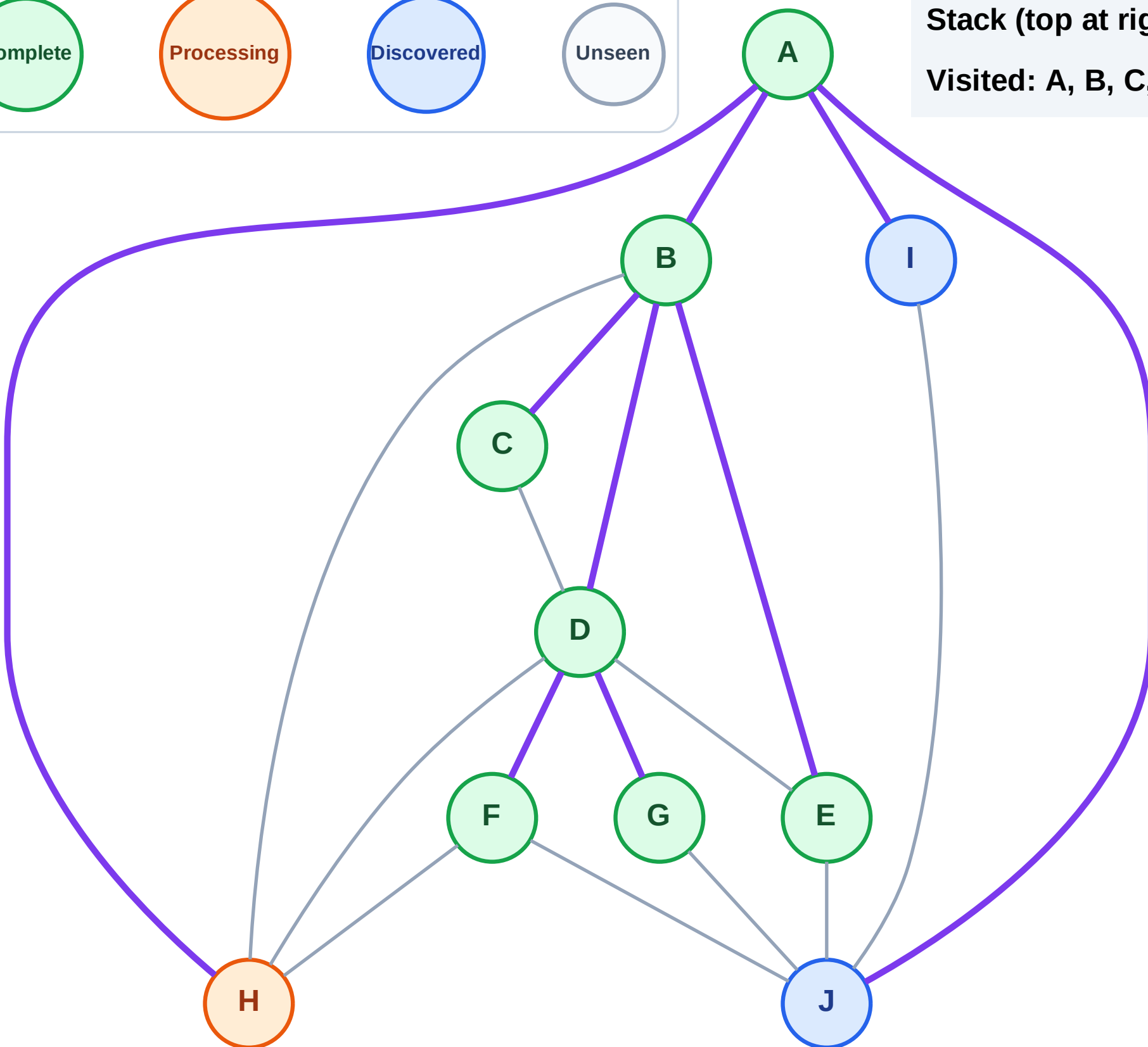
DFS Traversal

Step 16: Process node H

Legend



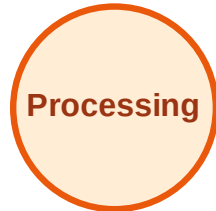
Stack (top at right): J | I
Visited: A, B, C, D, E, F, G



DFS Traversal

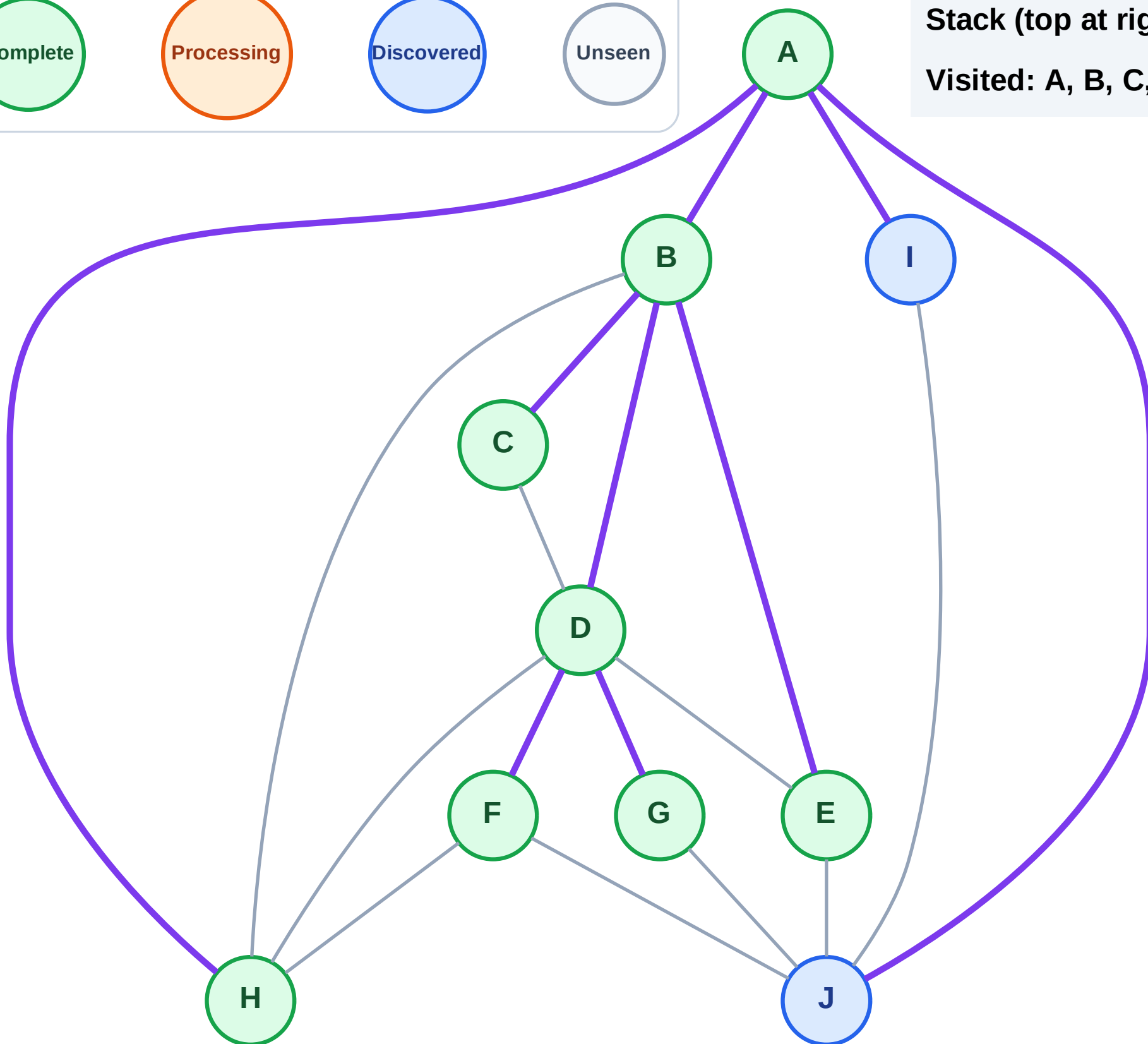
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

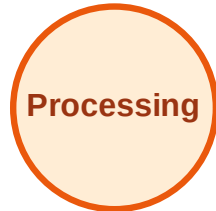
Visited: A, B, C, D, E, F, G, H



DFS Traversal

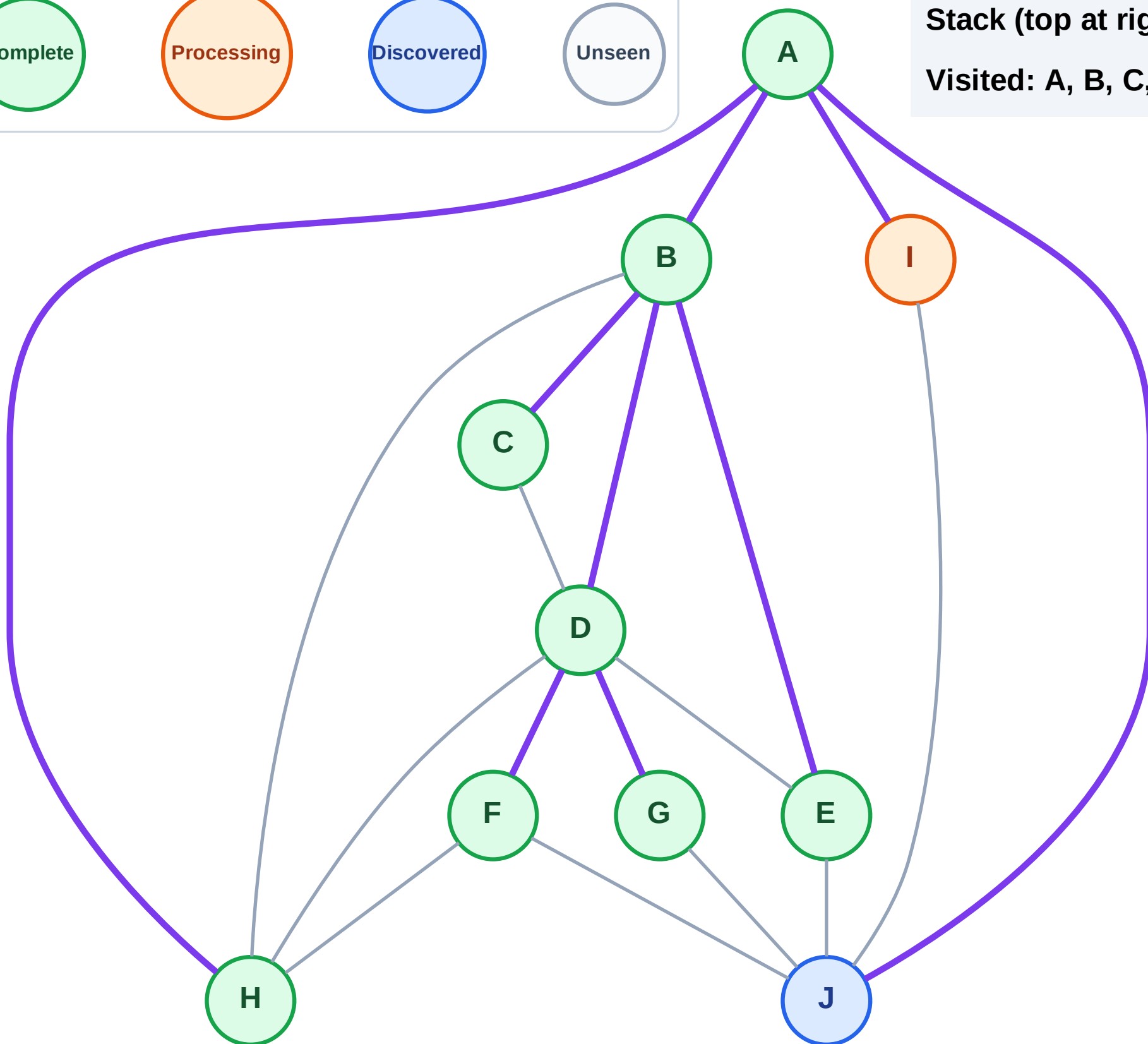
Step 18: Process node I

Legend



Stack (top at right): J

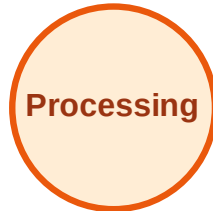
Visited: A, B, C, D, E, F, G, H



DFS Traversal

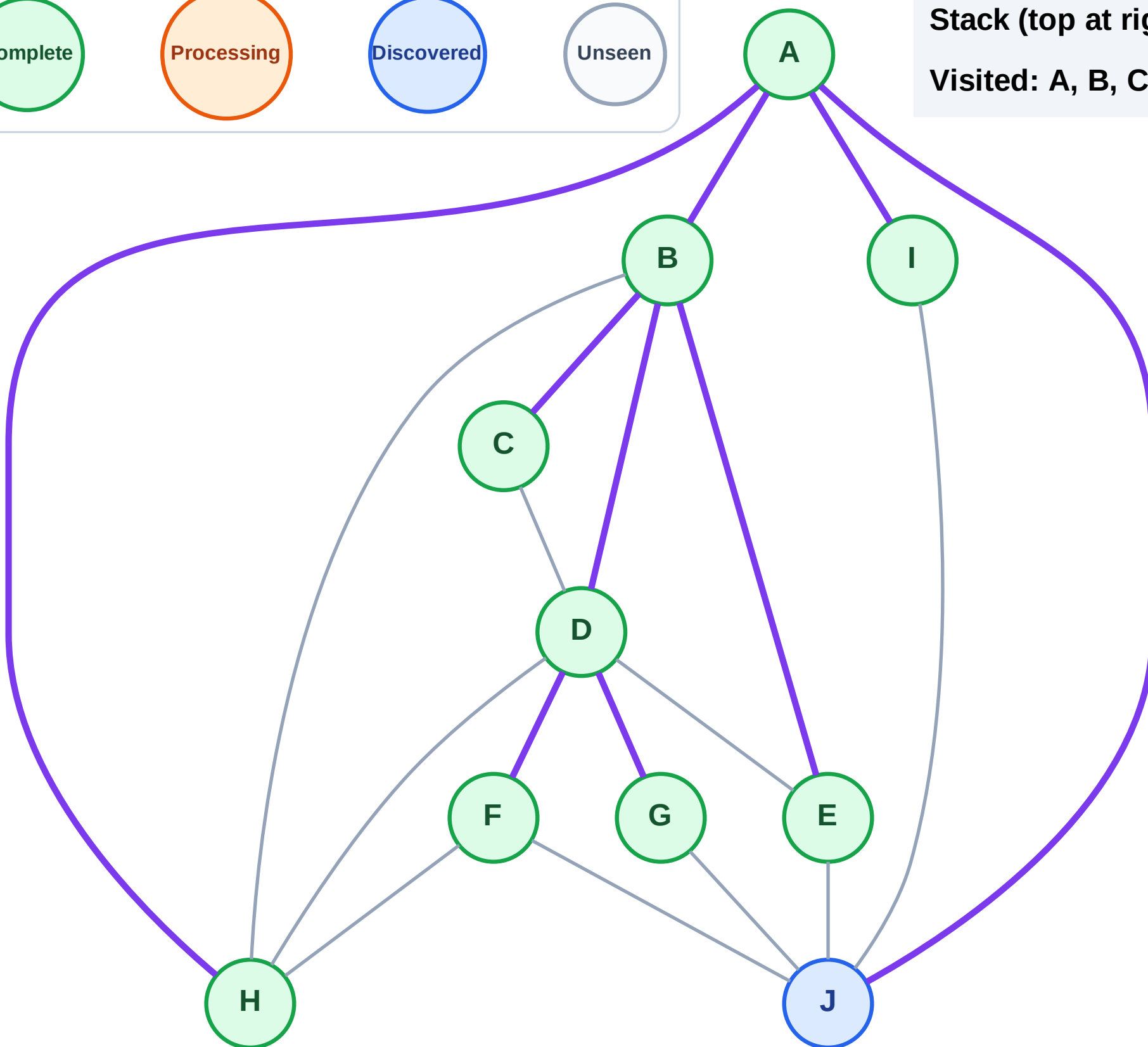
Step 19: No new neighbors from I

Legend



Stack (top at right): J

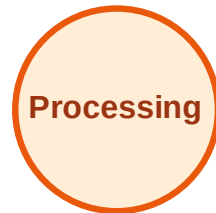
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

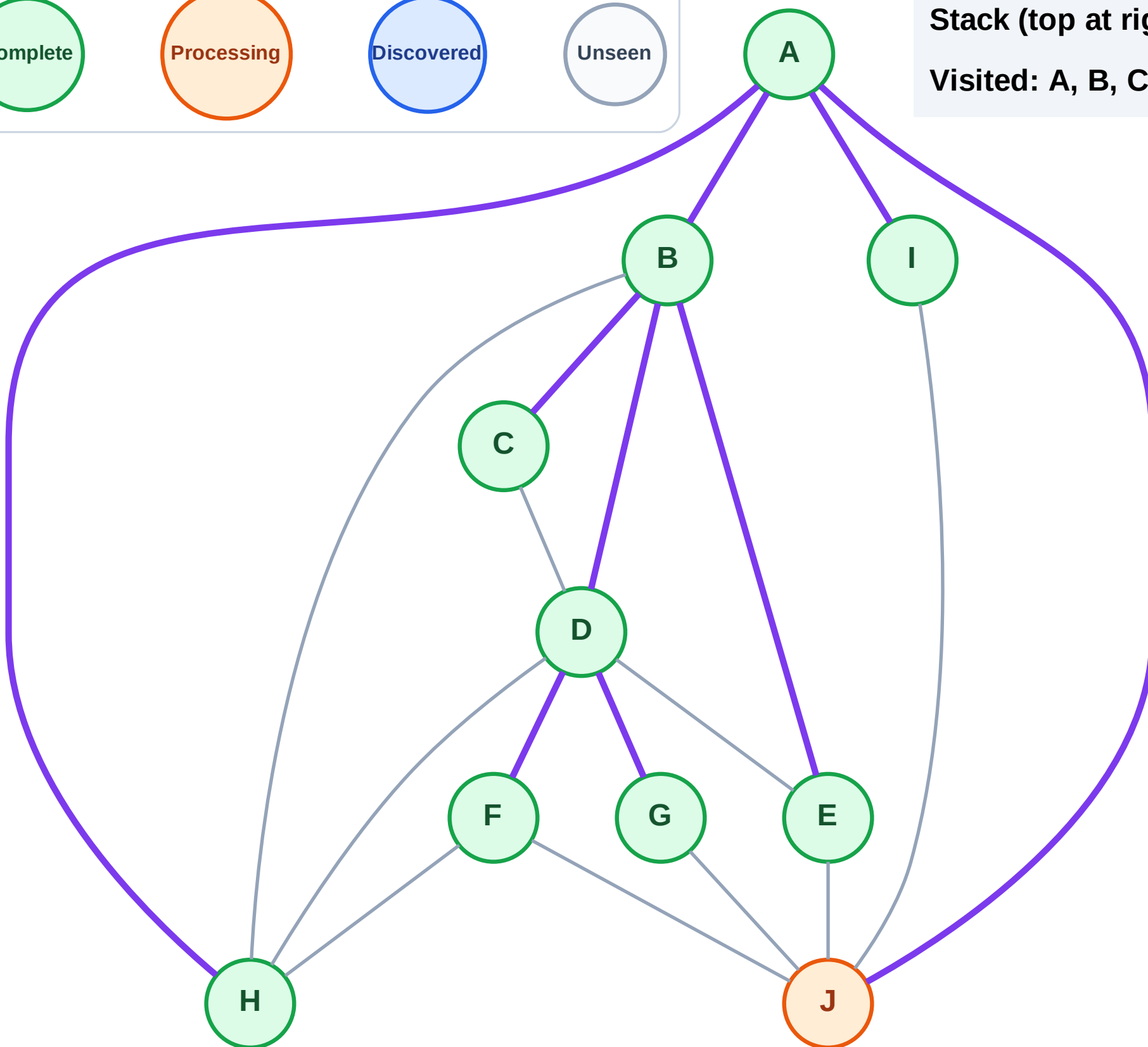
Step 20: Process node J

Legend



Stack (top at right): (empty)

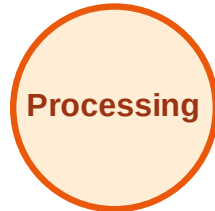
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

